Supplementary Information

Indirect reciprocity with inertial reputation updating

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This Supplementary Information gives the full derivations behind the Methods. It treats private assessment and the discrete inertial (direct-reset) reputation record used throughout the paper. Supplementary Notes 1–6 derive the model and its reputation dynamics, Notes 7–9 contain the invasion analyses, and Notes 10–13 describe simulations, numerical methods, the settings of every figure and the scope of the results. Figure numbers refer to the main text and Extended Data.

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Supplementary Note 1. Notation and parameters

Supplementary Table S1 lists every symbol used in the Methods and in this Supplementary Information. Each symbol has a single meaning. Two conventions should be noted. (i) The letters G and B name both the two values of a binary image (good, bad) and the two established latent states. Because an established state is always read as the image of the same name ($\sigma(\mathrm{G})=1$, $\sigma(\mathrm{B})=0$), this use is unambiguous: latent states appear only as arguments of $\sigma(\cdot)$, $\pi_{\cdot}$ and the transition rules, whereas images appear only as conditions in $P^{+}_{A|Y}$ and in the pair probabilities $h_{\mathrm{GG}}, h_{\mathrm{GB}}, h_{\mathrm{BB}}$. (ii) Subscripts C and D denote intended actions. An individual that always intends C (ALLC) or D (ALLD) is assessed positively with probability Φ_{C} or Φ_{D}; strategy-level quantities carry full strategy names (r_{ALLC}, u_{DISC}, ...). The subscript R denotes the resident of a homogeneous DISC population and X a rare mutant.

Table S1. Symbols.

Symbol	Meaning
Game, strategies and norms	
b, c	Benefit to the recipient and cost to the donor of realized cooperation; $b > c > 0$.
$I, J \in \{\mathrm{ALLC}, \mathrm{ALLD}, \mathrm{DISC}\}$	Strategies; f_I frequency, $\boldsymbol{f}$ composition, $\sum_I f_I = 1$.
$A \in \{\mathrm{C}, \mathrm{D}\}$	Intended action (cooperate, defect).
$Y \in \{\mathrm{G}, \mathrm{B}\}$	Binary image of a recipient held by an observer (good, bad).
ν_{AY}	Probability that the norm prescribes a positive assessment of realized action A towards a recipient with image Y: $\nu_{\mathrm{CG}}=1$, $\nu_{\mathrm{DG}}=0$, $\nu_{\mathrm{CB}}=p$, $\nu_{\mathrm{DB}}=q$.
p, q	Norm parameters; SC $(1,0)$, SH $(0,0)$, SJ $(0,1)$, SS $(1,1)$.
e_1, e_2	Implementation error (intended C becomes D) and assessment error (recorded assessment reversed).
ε, κ	$\varepsilon = (1-e_1)(1-e_2)+e_1e_2$; $\kappa = (1-e_1)(1-2e_2) = \varepsilon - e_2$.
$P^{+}_{A\|Y}$	Probability of a positive recorded assessment given intended action A and the observer's image Y of the recipient (errors included).
$\Phi_{\mathrm{C}}(r)$, $\Phi_{\mathrm{D}}(r)$	Probability that intended C or D is assessed positively when the observer regards the recipient as good with probability r.
$\Delta\Phi(r)$	Assessment contrast $\Phi_{\mathrm{C}}(r) - \Phi_{\mathrm{D}}(r) = \kappa[r + (p-q)(1-r)]$.
Inertial record	
$L_{ij} \in \mathscr{L}$	Latent record of observer i about target j; $\mathscr{L} = \{\mathrm{G}, \mathrm{W}_1, \ldots, \mathrm{W}_{k_1}, \mathrm{B}, \mathrm{S}_1, \ldots, \mathrm{S}_{k_2}\}$.
W_m, S_n	Warning states ($m = 1, \ldots, k_1$) and recovery states ($n = 1, \ldots, k_2$).
k_1, k_2, K	Tolerance depth, strictness depth, total depth $K = k_1 + k_2$.
α, β	Tolerance strength and strictness strength (readout parameters), $0 \le \alpha, \beta \le 1$.
$\sigma(\ell)$, $\boldsymbol{\sigma}$	Probability that state ℓ is read as good; column vector of these probabilities.
u, v	Probability that a recorded assessment of a target is positive; $v = 1-u$.
$T^{+}, T^{-}, T(u)$	Transition matrices after a positive or negative assessment; $T(u) = uT^{+} + vT^{-}$.
$\pi_\ell(u)$, $Z(u)$	Stationary distribution of a record with i.i.d. input u; normalization $Z = v^{k_1} + u^{k_2} - v^{k_1}u^{k_2}$.
$\mathscr{R}(u)$	Reputation map $\mathscr{R}(u;\alpha,\beta,k_1,k_2) = \sum_\ell \pi_\ell(u)\sigma(\ell)$; $\mathscr{R}' = \mathrm{d}\mathscr{R}/\mathrm{d}u$.
Σ_{W}, Σ_{S}, Ω	Warning-path weight, recovery-path weight and numerator of $\mathscr{R}$ (Supplementary Note 4).
$\Theta_{\mathrm{G}\to\mathrm{B}}$, $\Theta_{\mathrm{B}\to\mathrm{G}}$	Number of assessments needed to leave good (bad) standing, starting from G (B).
Population reputations	
r_I, $\bar{r}$	Probability that an observer regards a type-I individual as good; $\bar{r} = \sum_I f_I r_I$.
$h_{\mathrm{GG}}, h_{\mathrm{GB}}, h_{\mathrm{BB}}$	Probabilities that donor and observer both regard a random recipient as good, hold a given ordered disagreement, or both regard it as bad.
u_I	Probability that a type-I donor is assessed positively.
$\boldsymbol{\pi}_I$	Row vector of latent-state probabilities of records about type-I targets.
τ, t', t	Auxiliary relaxation time; time in assessments per record; evolutionary time.
$u_{\mathrm{R}}(r)$, $F(r)$	Resident assessment probability $r\Phi_{\mathrm{C}}(r) + (1-r)\Phi_{\mathrm{D}}(r)$ in a homogeneous DISC population; $F(r) = \mathscr{R}[u_{\mathrm{R}}(r)] - r$.

Symbol	Meaning (continued)
r_{R}^*, u_{R}^*, Φ_A^*, $\Delta\Phi^*$	Selected resident reputation (highest strictly stable root); $u_{\mathrm{R}}(r_{\mathrm{R}}^*)$; $\Phi_A(r_{\mathrm{R}}^*)$; $\Delta\Phi(r_{\mathrm{R}}^*)$.
Evolution and cooperation	
Π_I, $\bar{\Pi}$	Expected payoff of strategy I; population mean payoff.
ρ_{int}, ρ	Intended and realized cooperation rate, $\rho = (1-e_1)\rho_{\mathrm{int}}$.
$\bar{\rho}$, ω_{C}, $\bar{\rho}_{\mathrm{C}}$	Mean long-run cooperation over initial compositions; cooperative-basin size; cooperation within the basin.
n_{init}	Number of initial compositions (512 per sample).
Invasion	
χ_{X}	Probability that a rare mutant X intends to help a resident recipient.
u_{X}, r_{X}^*	Positive-assessment probability and stationary reputation of the mutant.
Λ_{X}	Normalized payoff advantage $(\Pi_{\mathrm{X}} - \Pi_{\mathrm{R}})/[c(1-e_1)]$.
λ_{ALLD}, λ_{ALLC}	Unnormalized payoff advantages of rare ALLD and ALLC over residents.
r_{ALLD}^*, r_{ALLC}^*	Reputations of rare ALLD and ALLC in the resident population, $\mathscr{R}(\Phi_{\mathrm{D}}^*)$ and $\mathscr{R}(\Phi_{\mathrm{C}}^*)$.
R_{L}, R_{U}	Lower and upper benefit-to-cost thresholds for resistance to ALLD and ALLC.
δ	Probability that O-DISC withholds an intended cooperation.
$\Lambda(\delta)$, $H(\delta)$	Normalized O-DISC advantage; reputation lost per unit of withheld cooperation.
$H_{\min}$, δ_{c}	$\min_{0\le\delta\le1} H(\delta)$; upper end of the interval with $\Lambda > 0$.
α_{L}, α_{U}	Lower (fold) and upper (invasion) boundaries of a secure readout interval.
$\mathscr{R}_{\mathrm{agg}}$, M, Q	Aggregation map; number of sampled assessments; required number of positive ones.
$\mathscr{R}_{\mathrm{score}}$, z, $z_{\max}$, z_0	Quantitative-score map; score; score bound; good-image threshold.
Finite-population simulations	
N, n_I	Population size and number of type-I individuals.
s_{ij}	$\sigma(L_{ij})$, the probability that DISC observer i currently reads target j as good.
g_j, g_I	Mean goodness of target j; average over type-I targets and sampling times.
d_j, d_I	Probability that two distinct DISC observers read target j differently; average over type-I targets and times.
Δg, $\bar{d}$	Reputational separation and mean disagreement.

Table S2. Reference parameters and inertia settings. Values apply unless a figure axis varies them.

Quantity	Value	Varied in
Benefit, cost	$b = 5$, $c = 1$	b/c: Extended Data Figs. 5, 9
Errors	$e_1 = e_2 = 0.02$	$e_1, e_2 \in [0, 0.45]$: Extended Data Fig. 6
No inertia	$(k_1, k_2) = (0,0)$; equivalently $\alpha = \beta = 0$	–
Full one-step inertia	$(k_1, k_2) = (1,1)$, $\alpha = \beta = 1$	(α, β): Extended Data Figs. 3, 8
Selected multi-step inertia	SC $(2,0)$, SH $(3,0)$, SJ $(4,3)$, SS $(1,2)$; $\alpha = \beta = 1$	α: Fig. 5, Extended Data Fig. 9
Depth scans	$\alpha = \beta = 1$	k_1, k_2: Figs. 2, 4; Extended Data Figs. 1, 3, 7
Initial records	all latent records in G ($r_I = 1$)	all-bad: Extended Data Fig. 2e

Supplementary Note 2. Interactions, social norms and errors

2.1 Sequence of an interaction

A donor, a recipient and an observer are drawn from the population, the observer independently of the two others. Each interaction proceeds as follows.

1. The donor forms an intended action. ALLC intends C, ALLD intends D, and DISC consults its own record of the recipient, samples a binary image and intends C if and only if the image is G.

2. Intended cooperation is implemented as defection with probability e_1; intended defection is always implemented. The recipient receives b and the donor pays c only if cooperation is realized.

3. The observer consults its own record of the recipient, samples a binary image Y and applies the norm to the realized action: a positive assessment is prescribed with probability ν_{AY}.

4. The prescribed assessment is reversed with probability e_2.

5. The recorded assessment updates the observer's record of the donor. No other record changes.

The donor's and the observer's images of the recipient come from different records, so the donor may act on a different view of the recipient from the one used to judge it. This is the defining feature of private assessment.

2.2 Error-integrated assessment probabilities

Conditional on a realized action A and an image Y, the recorded assessment is positive with probability $\nu_{AY}(1-e_2)+(1-\nu_{AY})e_2 = e_2+(1-2e_2)\nu_{AY}$. Intended cooperation is realized as C with probability $1-e_1$ and as D with probability e_1, whereas intended defection is always realized as D. Averaging over the realized action,

$$P^+_{\mathrm{C}|Y} = e_2+(1-2e_2)\big[(1-e_1)\nu_{\mathrm{C}Y}+e_1\nu_{\mathrm{D}Y}\big], \qquad P^+_{\mathrm{D}|Y} = e_2+(1-2e_2)\nu_{\mathrm{D}Y}. \tag{S1}$$

With $\nu_{\mathrm{CG}}=1$, $\nu_{\mathrm{DG}}=0$, $\nu_{\mathrm{CB}}=p$ and $\nu_{\mathrm{DB}}=q$,

$$P^+_{\mathrm{C}|\mathrm{G}} = \varepsilon, \qquad P^+_{\mathrm{D}|\mathrm{G}} = e_2, \qquad P^+_{\mathrm{C}|\mathrm{B}} = e_2+(1-2e_2)\big[(1-e_1)p+e_1q\big], \qquad P^+_{\mathrm{D}|\mathrm{B}} = e_2+(1-2e_2)q, \tag{S2}$$

where $\varepsilon = e_2+\kappa = (1-e_1)(1-e_2)+e_1e_2$ and $\kappa = (1-e_1)(1-2e_2)$. Equivalently, $P^+_{\mathrm{C}|\mathrm{B}} = p(\varepsilon-e_2)+q(1-\varepsilon-e_2)+e_2$, because $\varepsilon-e_2 = (1-e_1)(1-2e_2)$ and $1-\varepsilon-e_2 = e_1(1-2e_2)$. Supplementary Table S3 gives the four norms. These probabilities already include implementation error; the separate factor $1-e_1$ in the payoffs accounts for realized donations, not for assessments.

Table S3. Norms and error-integrated assessment probabilities.

Norm	$p=\nu_{\mathrm{CB}}$	$q=\nu_{\mathrm{DB}}$	$P^+_{\mathrm{C}\|\mathrm{G}}$	$P^+_{\mathrm{D}\|\mathrm{G}}$	$P^+_{\mathrm{C}\|\mathrm{B}}$	$P^+_{\mathrm{D}\|\mathrm{B}}$
Scoring (SC)	1	0	ε	e_2	ε	e_2
Shunning (SH)	0	0	ε	e_2	e_2	e_2
Stern Judging (SJ)	0	1	ε	e_2	$1-\varepsilon$	$1-e_2$
Simple Standing (SS)	1	1	ε	e_2	$1-e_2$	$1-e_2$

2.3 Action-conditional assessment probabilities

Suppose that the observer regards the recipient as good with probability r, independently of the donor's image. Then intended cooperation and intended defection are assessed positively with probabilities

$$\Phi_{\mathrm{C}}(r) = rP^+_{\mathrm{C}|\mathrm{G}}+(1-r)P^+_{\mathrm{C}|\mathrm{B}}, \qquad \Phi_{\mathrm{D}}(r) = rP^+_{\mathrm{D}|\mathrm{G}}+(1-r)P^+_{\mathrm{D}|\mathrm{B}}. \tag{S3}$$

Both are linear in r, with slopes $\Phi'_{\mathrm{C}} = P^+_{\mathrm{C}|\mathrm{G}}-P^+_{\mathrm{C}|\mathrm{B}}$ and $\Phi'_{\mathrm{D}} = P^+_{\mathrm{D}|\mathrm{G}}-P^+_{\mathrm{D}|\mathrm{B}}$. Their difference is

$$\Delta\Phi(r) = r(P^+_{\mathrm{C}|\mathrm{G}}-P^+_{\mathrm{D}|\mathrm{G}})+(1-r)(P^+_{\mathrm{C}|\mathrm{B}}-P^+_{\mathrm{D}|\mathrm{B}}) = \kappa[r+(p-q)(1-r)], \tag{S4}$$

because $P^+_{\mathrm{C}|\mathrm{G}}-P^+_{\mathrm{D}|\mathrm{G}} = \kappa$ and $P^+_{\mathrm{C}|\mathrm{B}}-P^+_{\mathrm{D}|\mathrm{B}} = (p-q)\kappa$. Thus $\Delta\Phi = \kappa$ (SC), κr (SH, SS) and $\kappa(2r-1)$ (SJ): under SJ the contrast changes sign at $r=1/2$, so on a low-reputation branch defection is assessed better than cooperation.

A DISC donor in a homogeneous DISC population with reputation r regards the recipient as good with probability r, independently of the observer (Supplementary Note 5). It therefore intends C with probability r and is assessed positively with probability

$$u_{\mathrm{R}}(r) = r\,\Phi_{\mathrm{C}}(r)+(1-r)\,\Phi_{\mathrm{D}}(r) = r^2P^+_{\mathrm{C}|\mathrm{G}}+r(1-r)\big(P^+_{\mathrm{C}|\mathrm{B}}+P^+_{\mathrm{D}|\mathrm{G}}\big)+(1-r)^2P^+_{\mathrm{D}|\mathrm{B}}, \tag{S5}$$

with derivative $u'_{\mathrm{R}}(r) = \Delta\Phi(r)+r\Phi'_{\mathrm{C}}+(1-r)\Phi'_{\mathrm{D}}$. Supplementary Table S4 lists these functions.

Table S4. Resident assessment functions. $\kappa = (1-e_1)(1-2e_2)$. For $e_1 = e_2 = 0.02$: $\varepsilon = 0.9608$, $\kappa = 0.9408$.

Norm	$\Phi_{\mathrm{C}}(r)$	$\Phi_{\mathrm{D}}(r)$	$\Delta\Phi(r)$	$u_{\mathrm{R}}(r)$
SC	ε	e_2	κ	$e_2+\kappa r$
SH	$e_2+\kappa r$	e_2	κr	$e_2+\kappa r^2$
SJ	$1-\varepsilon-(1-2\varepsilon)r$	$1-e_2-(1-2e_2)r$	$\kappa(2r-1)$	$1-e_2-(1-2e_2+\kappa)r+2\kappa r^2$
SS	$1-e_2-(1-e_2-\varepsilon)r$	$1-e_2-(1-2e_2)r$	κr	$1-e_2-(1-2e_2)r+\kappa r^2$

Supplementary Note 3. The inertial reputation record

3.1 States and transitions

Each observer holds, for each target, a latent record in $\mathscr{L} = \{\mathrm{G}, \mathrm{W}_1, \ldots, \mathrm{W}_{k_1}, \mathrm{B}, \mathrm{S}_1, \ldots, \mathrm{S}_{k_2}\}$ with $k_1, k_2 \in \{0, 1, 2, \ldots\}$. The established states G and B represent good and bad standing. Warning states W_m count consecutive negative assessments that challenge good standing, and recovery states S_n count consecutive positive assessments that challenge bad standing. Supplementary Table S5 lists the transitions. Two rules generate it: evidence against the established judgement advances along the corresponding path, and evidence supporting it resets the whole path (direct reset). The record has $K+2$ states. Consecutiveness refers to the assessments made by one observer about one target; these need not be consecutive interactions of the target.

Table S5. Transitions of a latent record. Rows for W_m and S_n are absent when $k_1 = 0$ or $k_2 = 0$, respectively.

Current state	After a positive assessment	After a negative assessment
G	G	W_1 if $k_1 \geq 1$; B if $k_1 = 0$
W_m, $1 \leq m < k_1$	G	W_{m+1}
W_{k_1}	G	B
B	S_1 if $k_2 \geq 1$; G if $k_2 = 0$	B
S_n, $1 \leq n < k_2$	S_{n+1}	B
S_{k_2}	G	B

We order the states as $(\mathrm{G}, \mathrm{W}_1, \ldots, \mathrm{W}_{k_1}, \mathrm{B}, \mathrm{S}_1, \ldots, \mathrm{S}_{k_2})$ and write T^+ and T^- for the $(K+2)\times(K+2)$ matrices with entry $[T^{\pm}]_{\ell\ell'} = 1$ if a positive (negative) assessment moves ℓ to ℓ' and 0 otherwise. Each row of $T^{\pm}$ contains exactly one 1. If a record receives a positive assessment with probability u and a negative one with probability $v = 1-u$, its state follows the Markov chain with transition matrix

$$T(u) = uT^+ + vT^-. \tag{S6}$$

For one-step inertia, in the order $(\mathrm{G}, \mathrm{W}_1, \mathrm{B}, \mathrm{S}_1)$,

$$T(u) = \begin{pmatrix} u & v & 0 & 0 \\ u & 0 & v & 0 \\ 0 & 0 & v & u \\ u & 0 & v & 0 \end{pmatrix}.$$

3.2 Binary readout

When a record in state ℓ is consulted, it yields the image G with probability $\sigma(\ell)$ and B otherwise, where

$$\sigma(\mathrm{G}) = 1, \qquad \sigma(\mathrm{W}_m) = \alpha^m, \qquad \sigma(\mathrm{B}) = 0, \qquad \sigma(\mathrm{S}_n) = 1-\beta^n, \qquad 0 \leq \alpha, \beta \leq 1. \tag{S7}$$

Each consultation draws a fresh image, independently of all other consultations given the current state, and does not change the state. In particular, when a DISC donor is also the observer of its own action, the image used to act and the image used to judge are independent draws from the same record. The geometric form makes successive warning states progressively less likely to be treated as good, and successive recovery states progressively more likely to be treated as good. The depths control how much consistent evidence completes a reversal of the latent state; the readout parameters control how strongly incomplete evidence already affects behaviour and assessment.

3.3 Settings that reduce to instantaneous updating

Lemma S1. *If $(k_1,k_2)=(0,0)$, or if $\alpha=\beta=0$ at any depths, the image of a record that has received at least one assessment equals its most recent assessment, and a record that has received none yields* G.

Proof. For $(k_1,k_2)=(0,0)$ the record has the states G and B only, and every assessment sets it to G or B according to its sign. For $\alpha=\beta=0$ and any depths, the states that can follow a positive assessment are G and $\mathrm{S}_1,\ldots,\mathrm{S}_{k_2}$, all with $\sigma=1$ because $1-0^n=1$ for $n\geq 1$; the states that can follow a negative assessment are $\mathrm{W}_1,\ldots,\mathrm{W}_{k_1}$ and B, all with $\sigma=0$ because $0^m=0$ for $m\geq 1$. Initially every record is in G. □

The two implementations therefore generate identical image processes, not only identical stationary statistics. We use $(0,0)$ in the depth scans and in the stationary analyses, and the one-step state space with $\alpha=\beta=0$ in the instantaneous panels of Figs. 2a–d and 3.

Supplementary Note 4. The stationary reputation map

4.1 Stationary distribution

Proposition S2 (Stationary distribution)**.** *For $0<u<1$ the chain $T(u)$ is irreducible and aperiodic, and its unique stationary distribution is*

$$\pi_{\mathrm{G}}=\frac{u^{k_2+1}}{Z},\qquad \pi_{\mathrm{W}_m}=v^m\pi_{\mathrm{G}},\qquad \pi_{\mathrm{B}}=\frac{v^{k_1+1}}{Z},\qquad \pi_{\mathrm{S}_n}=u^n\pi_{\mathrm{B}},\qquad Z(u)=v^{k_1}+u^{k_2}-v^{k_1}u^{k_2}, \tag{S8}$$

for $1\leq m\leq k_1$ and $1\leq n\leq k_2$.

Proof. Irreducibility: from any state, a run of negative assessments reaches B and a run of positive assessments then reaches G; from G every warning state and from B every recovery state is reachable. Aperiodicity: G has a self-loop with probability $u>0$. Hence a unique stationary distribution exists.

Paths. The state W_m ($m\geq 1$) is entered only from W_{m-1}, with $\mathrm{W}_0\equiv\mathrm{G}$, after a negative assessment. Stationarity of W_m requires $\pi_{\mathrm{W}_m}=v\,\pi_{\mathrm{W}_{m-1}}$, hence $\pi_{\mathrm{W}_m}=v^m\pi_{\mathrm{G}}$. Likewise, S_n is entered only from S_{n-1}, with $\mathrm{S}_0\equiv\mathrm{B}$, after a positive assessment, so $\pi_{\mathrm{S}_n}=u^n\pi_{\mathrm{B}}$.

Flux balance. Partition $\mathscr{L}$ into the good-standing block $\{\mathrm{G},\mathrm{W}_1,\ldots,\mathrm{W}_{k_1}\}$ and the bad-standing block $\{\mathrm{B},\mathrm{S}_1,\ldots,\mathrm{S}_{k_2}\}$. The only transition from the first block to the second is the negative assessment from W_{k_1} (from G if $k_1=0$), with stationary flux $v\,\pi_{\mathrm{W}_{k_1}}=v^{k_1+1}\pi_{\mathrm{G}}$. The only transition in the opposite direction is the positive assessment from S_{k_2} (from B if $k_2=0$), with flux $u^{k_2+1}\pi_{\mathrm{B}}$. In a stationary state the two fluxes are equal:

$$v^{k_1+1}\pi_{\mathrm{G}}=u^{k_2+1}\pi_{\mathrm{B}}. \tag{S9}$$

Normalization. Writing $\pi_{\mathrm{G}}=u^{k_2+1}/Z$ and $\pi_{\mathrm{B}}=v^{k_1+1}/Z$ satisfies Eq. (S9). Summing all probabilities,

$$Z=u^{k_2+1}\sum_{m=0}^{k_1}v^m+v^{k_1+1}\sum_{n=0}^{k_2}u^n=u^{k_2}\left(1-v^{k_1+1}\right)+v^{k_1}\left(1-u^{k_2+1}\right)=v^{k_1}+u^{k_2}-v^{k_1}u^{k_2}(u+v),$$

which is the stated Z because $u+v=1$. □

The exponents record the evidence requirements: completing a reversal of good standing requires k_1+1 negative assessments in a row, which suppresses π_{B} by v^{k_1+1}, and completing a recovery requires k_2+1 positive assessments in a row, which suppresses π_{G} by u^{k_2+1}.

4.2 The reputation map

Averaging the readout probabilities over Eq. (S8) gives the probability that a target is regarded as good in the stationary state of a record with i.i.d. input u:

$$\mathscr{R}(u) \equiv \mathscr{R}(u;\alpha,\beta,k_1,k_2) = \pi_{\mathrm{G}} + \sum_{m=1}^{k_1} \alpha^m \pi_{\mathrm{W}_m} + \sum_{n=1}^{k_2} (1-\beta^n)\pi_{\mathrm{S}_n} = \frac{u^{k_2+1}\,\Sigma_{\mathrm{W}}(v) + v^{k_1+1}\,\Sigma_{\mathrm{S}}(u)}{Z(u)}, \tag{S10}$$

$$\Sigma_{\mathrm{W}}(v) = \sum_{m=0}^{k_1} (\alpha v)^m, \qquad \Sigma_{\mathrm{S}}(u) = \sum_{n=1}^{k_2} u^n(1-\beta^n).$$

Empty sums vanish. At the endpoints, the chain is absorbed in B ($u=0$) or G ($u=1$), so $\mathscr{R}(0)=0$ and $\mathscr{R}(1)=1$; Eq. (S10) attains these values by continuity (for $k_1=0$ or $k_2=0$ it is already regular there). $\mathscr{R}(u)$ is exact for a record with i.i.d. input; how u is determined in a population is the subject of Supplementary Note 5.

Special cases.

1. No inertia. For $(k_1,k_2)=(0,0)$, $Z=1$ and $\mathscr{R}(u)=u$. By Lemma S1, the same holds for $\alpha=\beta=0$ at any depths.
2. One-step inertia, $(k_1,k_2)=(1,1)$:

$$\mathscr{R}(u) = \frac{u^2\,[1+\alpha(1-u)] + (1-\beta)\,u(1-u)^2}{1-u+u^2}, \qquad \mathscr{R}(u)-u = \frac{u(1-u)}{1-u+u^2}\,\big[\alpha u - \beta(1-u)\big]. \tag{S11}$$

   Tolerance raises and strictness lowers the stationary reputation relative to instantaneous updating.

3. Full readout, $\alpha=\beta=1$: $\Sigma_{\mathrm{W}} = (1-v^{k_1+1})/u$ and $\Sigma_{\mathrm{S}}=0$, so

$$\mathscr{R}(u) = \frac{u^{k_2}\big(1-v^{k_1+1}\big)}{Z(u)}, \qquad 1-\mathscr{R}(u) = \frac{v^{k_1}\big(1-u^{k_2+1}\big)}{Z(u)}. \tag{S12}$$

   Exchanging $u \leftrightarrow v$ and $k_1 \leftrightarrow k_2$ exchanges $\mathscr{R} \leftrightarrow 1-\mathscr{R}$: tolerance and strictness depths act symmetrically on the two kinds of reputational error.

4. No recovery path, $k_2=0$: $Z=1$ and $\mathscr{R}(u) = u\sum_{m=0}^{k_1}(\alpha v)^m$; β has no effect.

4.3 Interpretation through passage times

Let $\Theta_{\mathrm{G}\to\mathrm{B}}$ be the number of assessments a record starting in G receives until it first enters B, and $\Theta_{\mathrm{B}\to\mathrm{G}}$ the number from B until it first enters G. Leaving good standing requires a run of k_1+1 negative assessments, and leaving bad standing a run of k_2+1 positive ones. The classical waiting time for a run gives

$$\mathbb{E}[\Theta_{\mathrm{G}\to\mathrm{B}}] = \frac{1-v^{k_1+1}}{u\,v^{k_1+1}}, \qquad \mathbb{E}[\Theta_{\mathrm{B}\to\mathrm{G}}] = \frac{1-u^{k_2+1}}{v\,u^{k_2+1}}. \tag{S13}$$

The good-standing block is entered only at G and the bad-standing block only at B, so the record alternates between sojourns of these two kinds. Renewal–reward then gives the stationary probability of good standing, $\mathbb{E}[\Theta_{\mathrm{G}\to\mathrm{B}}]/(\mathbb{E}[\Theta_{\mathrm{G}\to\mathrm{B}}]+\mathbb{E}[\Theta_{\mathrm{B}\to\mathrm{G}}])$, which coincides with Eq. (S12). With full readout, therefore, $\mathscr{R}(u)$ is the fraction of time spent in good standing, and the depths enter only through the two mean sojourn times. With partial readout ($\alpha<1$ or $\beta<1$), part of each sojourn is read as the opposite image.

4.4 Properties used in the analysis

Proposition S3. *Let* $0<u<1$.

(a) Readout monotonicity:

$$\partial_\alpha \mathscr{R} = \frac{u^{k_2+1}}{Z}\sum_{m=1}^{k_1} m\,\alpha^{m-1}v^m \;\geq 0, \qquad \partial_\beta \mathscr{R} = -\frac{v^{k_1+1}}{Z}\sum_{n=1}^{k_2} n\,\beta^{n-1}u^n \;\leq 0. \tag{S14}$$

(b) Derivative. With $\Omega = u^{k_2+1}\Sigma_W + v^{k_1+1}\Sigma_S$ the numerator of Eq. (S10),

$$\mathscr{R}' = \frac{\Omega' Z - \Omega Z'}{Z^2}, \quad \begin{aligned} \Omega' &= (k_2+1)u^{k_2}\Sigma_W - u^{k_2+1}\textstyle\sum_{m=1}^{k_1} m\alpha^m v^{m-1} - (k_1+1)v^{k_1}\Sigma_S + v^{k_1+1}\sum_{n=1}^{k_2} n u^{n-1}(1-\beta^n), \\ Z' &= -k_1 v^{k_1-1}(1-u^{k_2}) + k_2 u^{k_2-1}(1-v^{k_1}). \end{aligned} \tag{S15}$$

(c) Response of a mostly good record ($v \to 0$). If $k_1 \geq 1$,

$$1 - \mathscr{R}(1-v) = (1-\alpha)\,v + O(v^2), \qquad \mathscr{R}'(1) = 1-\alpha. \tag{S16}$$

If $\alpha = 1$, for any $k_1 \geq 0$,

$$1 - \mathscr{R}(1-v) = \Big(1 + \sum_{n=1}^{k_2} \beta^n\Big) v^{k_1+1} + O(v^{k_1+2}), \qquad \mathscr{R}'(1-v) = (k_1+1)\Big(1 + \sum_{n=1}^{k_2} \beta^n\Big) v^{k_1} + O(v^{k_1+1}). \tag{S17}$$

(d) Response of a mostly bad record ($u \to 0$). If $k_2 \geq 1$, $\mathscr{R}(u) = (1-\beta)u + O(u^2)$. If $\beta = 1$, for any $k_2 \geq 0$, $\mathscr{R}(u) = \big(1 + \sum_{m=1}^{k_1} \alpha^m\big) u^{k_2+1} + O(u^{k_2+2})$.

Proof. (a) and (b) follow by differentiating Eq. (S10); Z does not depend on α or β. (c) Write $1 - \mathscr{R} = \pi_B + \sum_m (1-\alpha^m)\pi_{W_m} + \sum_n \beta^n \pi_{S_n}$. As $v \to 0$ with $k_1 \geq 1$, $Z \to 1$, $\pi_{W_1} = v\pi_G = v + O(v^2)$, and $\pi_B, \pi_{W_{m\geq 2}}, \pi_{S_n} = O(v^2)$, which gives the first statement. If $\alpha = 1$ the warning terms vanish, and $1 - \mathscr{R} = \pi_B\big(1 + \sum_n \beta^n u^n\big) = v^{k_1+1}\big(1 + \sum_n \beta^n\big) + O(v^{k_1+2})$; differentiating gives the derivative. (d) follows in the same way from $\mathscr{R} = \pi_G \Sigma_W + \sum_n (1-\beta^n)\pi_{S_n}$. □

Parts (c) and (d) quantify the two errors that inertia filters. A good individual that is occasionally assessed negatively (rate v) is read as bad with probability $(1-\alpha)v$ as soon as a warning path exists, irrespective of its length; only with $\alpha = 1$ does this error become of order v^{k_1+1}. Symmetrically, a bad individual that is occasionally assessed positively is read as good with probability $(1-\beta)u$, or of order u^{k_2+1} when $\beta = 1$. The same expansion implies that, near an all-good state, the marginal reputational penalty for an additional negative assessment is $\mathscr{R}'(1) = 1-\alpha$, which vanishes under full tolerance readout (Supplementary Note 8).

Proposition S4 (Concavity without a recovery path)**.** *If $k_2 = 0$, $\mathscr{R}$ is concave on $[0,1]$ for every $\alpha \in [0,1]$ and $k_1 \geq 0$.*

Proof. For $k_2 = 0$, $\mathscr{R}(u) = u\sum_{m=0}^{k_1} \alpha^m v^m$. Using $u\sum_{i=0}^{j} v^i = 1 - v^{j+1}$,

$$\mathscr{R}(u) = (1-\alpha) \sum_{j=0}^{k_1-1} \alpha^j \big[1 - v^{j+1}\big] + \alpha^{k_1}\big[1 - v^{k_1+1}\big],$$

because the coefficient of $u v^i$ on the right is $(1-\alpha)\sum_{j=i}^{k_1-1} \alpha^j + \alpha^{k_1} = \alpha^i$. The weights are non-negative, and each $1 - (1-u)^{j+1}$ is concave in u. □

Proposition S4 covers the selected SC $(2,0)$ and SH $(3,0)$ architectures. With a recovery path and $\beta = 1$, $\mathscr{R}$ is not concave: it starts as u^{k_2+1} near $u = 0$, and for $k_1 \geq 1$ and $\alpha = 1$ it approaches 1 as $1 - c v^{k_1+1}$ near $u = 1$.

Supplementary Note 5. Reputation dynamics in the population

5.1 Reputation quantities

At fixed strategy composition $\boldsymbol{f}$, let r_I be the probability that a randomly chosen observer regards a randomly chosen type-I individual as good, that is, the expected readout probability $\sigma(L_{ij})$ over observers i and type-I targets j. The probability that a random recipient is regarded as good is $\bar{r} = \sum_I f_I r_I$. Because all individuals use the same norm and record mechanism, all observers are statistically equivalent in the mean-field description; in the finite simulations, only DISC observers are stored because only they act on reputations (Supplementary Note 10).

5.2 Mean-field closure

Two assumptions turn the single-record result of Supplementary Note 4 into population equations; both are standard in analyses of private assessment [1, 2, 4, 5, 6].

(A1) *Conditional independence of images.* The images that the donor and the observer hold of a recipient are independent given the recipient's strategy.

(A2) *Independent inputs.* The successive assessments entering any record about a type-I target are independent, each positive with the same probability u_I.

(A1) neglects correlations between the records that different observers hold of the same individual. Such correlations arise because two observers may have assessed the same past actions of that individual, and because targets of the same strategy differ in their histories. (A2) neglects temporal correlations in the inputs to a record. The finite-population simulations retain both effects and are used to test reputation formation without the closure.

5.3 Positive-assessment probabilities

Under (A1), donor and observer both regard a random recipient as good, both as bad, or hold a given ordered disagreement with probabilities

$$h_{\mathrm{GG}} = \sum_J f_J r_J^2, \qquad h_{\mathrm{BB}} = \sum_J f_J (1-r_J)^2, \qquad h_{\mathrm{GB}} = \sum_J f_J r_J (1-r_J), \tag{S18}$$

with $h_{\mathrm{GG}} + 2h_{\mathrm{GB}} + h_{\mathrm{BB}} = 1$. Conditioning on the recipient's strategy matters: $h_{\mathrm{GG}} \geq \bar{r}^2$, with equality only if all strategies have the same reputation. An ALLC donor intends C whatever the recipient, and an ALLD donor intends D; their assessments depend only on the observer's image. A DISC donor intends C if its own image is G; Supplementary Table S6 lists the four cases. Summing,

$$u_{\mathrm{ALLC}} = \Phi_{\mathrm{C}}(\bar{r}), \qquad u_{\mathrm{ALLD}} = \Phi_{\mathrm{D}}(\bar{r}), \qquad u_{\mathrm{DISC}} = h_{\mathrm{GG}} P^+_{\mathrm{C|G}} + h_{\mathrm{GB}} \left(P^+_{\mathrm{C|B}} + P^+_{\mathrm{D|G}} \right) + h_{\mathrm{BB}} P^+_{\mathrm{D|B}}. \tag{S19}$$

Using Eq. (S5), the DISC expression can be written as $u_{\mathrm{DISC}} = \sum_J f_J u_{\mathrm{R}}(r_J)$: a DISC donor is assessed as if it were a resident in a homogeneous population whose reputation equals that of its recipient, averaged over recipients.

Table S6. Assessment of a DISC donor. The first image is the donor's, the second the observer's.

Images (donor, observer)	Probability	Intended action	Positive assessment
(G, G)	h_{GG}	C	$P^+_{\mathrm{C|G}}$
(G, B)	h_{GB}	C	$P^+_{\mathrm{C|B}}$
(B, G)	h_{GB}	D	$P^+_{\mathrm{D|G}}$
(B, B)	h_{BB}	D	$P^+_{\mathrm{D|B}}$

5.4 Self-consistency

Under (A2), each record about a type-I target has the stationary response $\mathscr{R}(u_I)$. Stationary reputations therefore satisfy

$$r_I^* = \mathscr{R}\big(u_I(\boldsymbol{r}^*; \boldsymbol{f})\big), \qquad I \in \{\mathrm{ALLC}, \mathrm{ALLD}, \mathrm{DISC}\}. \tag{S20}$$

Reputations determine assessments through Eq. (S19), and assessments determine reputations through $\mathscr{R}$. Because u_{ALLC} and u_{ALLD} depend on $\boldsymbol{r}$ only through $\bar{r}$, Eq. (S20) is a three-dimensional fixed-point problem that can have several solutions.

5.5 Full latent-state dynamics and the stationary reduction

Within the same closure, let $\boldsymbol{\pi}_I(t')$ be the row vector of latent-state probabilities of records about type-I targets, with time t' measured in assessments per record. Asynchronous updating, in which each record is updated at unit rate, gives

$$\frac{\mathrm{d}\boldsymbol{\pi}_I}{\mathrm{d}t'} = \boldsymbol{\pi}_I \big\{ T\left[u_I(\boldsymbol{r}; \boldsymbol{f})\right] - \mathbf{I} \big\}, \qquad r_I = \boldsymbol{\pi}_I \boldsymbol{\sigma}. \tag{S21}$$

This flow conserves probability and non-negativity. It is the mean-field limit of the finite simulation when time is measured in sweeps (one update per stored record on average) and assessment probabilities are replaced by their closed forms.

Proposition S5 (Stationary equivalence)**.** *Suppose $0 < u_I < 1$ at a stationary state. Then $\{\boldsymbol{\pi}_I^*\}$ is a stationary state of Eq. (S21) if and only if $\boldsymbol{\pi}_I^* = \boldsymbol{\pi}(u_I^*)$ with $u_I^* = u_I(\boldsymbol{r}^*; \boldsymbol{f})$ and $\boldsymbol{r}^*$ solves Eq. (S20).*

Proof. At a stationary state, $\boldsymbol{\pi}_I^* T(u_I^*) = \boldsymbol{\pi}_I^*$, so $\boldsymbol{\pi}_I^*$ is the unique stationary distribution of $T(u_I^*)$ (Proposition S2), and $r_I^* = \boldsymbol{\pi}_I^* \boldsymbol{\sigma} = \mathscr{R}(u_I^*)$. Conversely, given a solution $\boldsymbol{r}^*$ of Eq. (S20), the distributions $\boldsymbol{\pi}(u_I^*)$ are stationary and reproduce $\boldsymbol{r}^*$. □

Transient dynamics are not reduced in the same way. Projecting Eq. (S21) gives $\mathrm{d}r_I/\mathrm{d}t' = \boldsymbol{\pi}_I T(u_I)\boldsymbol{\sigma} - r_I$, and $\boldsymbol{\pi}_I T(u_I)\boldsymbol{\sigma}$ (the expected readout after one more update) generally differs from $\mathscr{R}(u_I)$. For example, for one-step inertia with $\alpha = \beta = 1$ and all records in G, $\boldsymbol{\pi}_I T(u)\boldsymbol{\sigma} = 1$ (both G and W_1 are read as good), whereas $\mathscr{R}(u) = u^2(2-u)/(1-u+u^2) < 1$. Two populations with the same r_I but different amounts of accumulated evidence evolve differently.

5.6 Auxiliary relaxation and branch selection

To select a stationary solution we use the auxiliary flow

$$\frac{\mathrm{d}r_I}{\mathrm{d}\tau} = \mathscr{R}\big(u_I(\boldsymbol{r}; \boldsymbol{f})\big) - r_I, \tag{S22}$$

whose fixed points are exactly the solutions of Eq. (S20). Its Jacobian is $\mathrm{diag}[\mathscr{R}'(u_I)]\,\partial \boldsymbol{u}/\partial \boldsymbol{r} - \mathbf{I}$. The time τ is a computational device; Eq. (S22) is not the transient dynamics of the records, and its stability need not coincide with that of Eq. (S21) in general. Three selection protocols are used.

1. *Main convention* (all strategy-dynamics results): at every composition, Eq. (S22) is integrated from $r_I(0) = 1$, which corresponds to all records in G. The replicator field is thus defined on the branch reached from good standing.

2. *Highest strictly stable root* (homogeneous DISC: Extended Data Figs. 2–3, local invasion and Fig. 5): the highest root of $F(r)$ with $F'(r) < 0$. It coincides with protocol 1 unless the highest root is a fold, which is marginal.

3. *Quasi-static tracking* (Extended Data Fig. 2e): reputations are initialized once, at $t = 0$, as all-good or all-bad and then follow the strategy trajectory, Eq. (S22) being re-solved from the current reputations at every evaluation. A branch is then left only where it ends at a fold, so the outcome can depend on reputational history.

5.7 Homogeneous discriminator populations

In a homogeneous DISC population, $h_{\mathrm{GG}} = r^2$, $h_{\mathrm{GB}} = r(1-r)$ and $h_{\mathrm{BB}} = (1-r)^2$, and Eq. (S19) reduces to Eq. (S5). The stationary resident reputations are the roots of

$$F(r) = \mathscr{R}\big[u_{\mathrm{R}}(r)\big] - r, \qquad F'(r) = \mathscr{R}'\big[u_{\mathrm{R}}(r)\big]\, u_{\mathrm{R}}'(r) - 1. \tag{S23}$$

A simple root is stable in the scalar flow $\mathrm{d}r/\mathrm{d}\tau = F(r)$ if $F' < 0$ and unstable if $F' > 0$. Because $F(0) \geq 0$ and $F(1) \leq 0$, roots alternate in stability, and a pair of roots appears or disappears at a fold, $F = F' = 0$. Bistability (Extended Data Figs. 2 and 3) means two stable roots separated by an unstable one; with errors, $F(1) < 0$, so the highest root is approached from $r = 1$. For every root in the full-readout depth planes ($0 \leq k_1, k_2 \leq 8$), the classification by F' agreed with the leading eigenvalue of Eq. (S21) linearized about $\boldsymbol{\pi}(u_{\mathrm{R}}(r))$ on the probability simplex; this linearization includes the dependence of u_{R} on the current readout.

Sensitivity to readout. On a regular stable branch ($F' < 0$), implicit differentiation of $F(r^*; \theta) = 0$ for $\theta \in \{\alpha, \beta\}$ gives

$$\frac{\partial r^*}{\partial \theta} = -\frac{\partial_\theta \mathscr{R}\big[u_{\mathrm{R}}(r^*)\big]}{F'(r^*)}, \tag{S24}$$

so by Eq. (S14) the resident reputation increases with α and decreases with β along a stable branch. The derivative diverges at a fold, where the branch ends.

5.8 The ALLD–DISC edge

On the edge $f_{\mathrm{ALLC}} = 0$, $f_{\mathrm{DISC}} = x$, $f_{\mathrm{ALLD}} = 1-x$, the three-dimensional problem reduces exactly to one equation. Given $\bar{r}$, $r_{\mathrm{ALLD}} = \mathscr{R}[\Phi_{\mathrm{D}}(\bar{r})]$ and $r_{\mathrm{ALLC}} = \mathscr{R}[\Phi_{\mathrm{C}}(\bar{r})]$ follow directly, and $r_{\mathrm{DISC}} = [\bar{r} - (1-x) r_{\mathrm{ALLD}}]/x$ follows from the definition of $\bar{r}$. Substituting these into $r_{\mathrm{DISC}} = \mathscr{R}\left[x\, u_{\mathrm{R}}(r_{\mathrm{DISC}}) + (1-x)\, u_{\mathrm{R}}(r_{\mathrm{ALLD}})\right]$, which is $r_{\mathrm{DISC}} = \mathscr{R}(u_{\mathrm{DISC}})$ written with $u_{\mathrm{DISC}} = \sum_J f_J\, u_{\mathrm{R}}(r_J)$, gives one equation in $\bar{r}$, whose roots with $r_{\mathrm{DISC}} \in [0,1]$ are all self-consistent states on the edge. Their stability is that of Eq. (S22). The normalized payoff advantage of DISC over ALLD on the edge is $(\Pi_{\mathrm{DISC}} - \Pi_{\mathrm{ALLD}})/[c(1-e_1)] = (b/c)\, x\, (r^*_{\mathrm{DISC}} - r^*_{\mathrm{ALLD}}) - \bar{r}^*$ (Supplementary Note 6). Where it changes sign continuously, the edge has an equilibrium; where the selected branch ends at a fold, the sign can change discontinuously (Extended Data Fig. 4).

Supplementary Note 6. Strategy dynamics and cooperation

6.1 Payoffs

Reputations are assumed to reach their stationary state quickly compared with changes in strategy frequencies, as in previous models of indirect reciprocity [2, 4, 5, 3]. Payoffs are computed per unit time, in which each individual acts on average once as donor and once as recipient. A type-I recipient receives intended help from ALLC donors (probability f_{ALLC}) and from DISC donors that regard it as good (probability $f_{\mathrm{DISC}} r^*_I$). As a donor, ALLC intends to help with probability 1, ALLD with probability 0, and DISC with probability $\bar{r}^*$, because its recipient is a random individual. Each intended donation is realized with probability $1-e_1$, so

$$\Pi_I = (1-e_1)\left[b\left(f_{\mathrm{ALLC}} + f_{\mathrm{DISC}} r^*_I\right) - c\,\chi_I\right], \qquad \chi_{\mathrm{ALLC}} = 1,\ \chi_{\mathrm{ALLD}} = 0,\ \chi_{\mathrm{DISC}} = \bar{r}^*, \tag{S25}$$

which gives the payoffs of the Methods. Every realized donation creates one benefit and one cost, so $\bar{\Pi} = \sum_I f_I \Pi_I = (b-c)\rho$, where $\rho = (1-e_1)\rho_{\mathrm{int}}$ and $\rho_{\mathrm{int}} = f_{\mathrm{ALLC}} + f_{\mathrm{DISC}} \bar{r}^*$. The factor $1-e_1$ rescales time in Eq. (S26), but e_1 also changes the reputations through $P^+_{A|Y}$.

6.2 Replicator dynamics and equilibria

Strategy frequencies follow

$$\frac{\mathrm{d}f_I}{\mathrm{d}t} = f_I\left(\Pi_I - \bar{\Pi}\right), \tag{S26}$$

with stationary reputations recomputed at every composition. Edges and vertices are invariant. At a vertex where all individuals use J, the eigenvalues are the growth rates $\Pi_I - \Pi_J$ of the two absent strategies, evaluated with the reputations of that vertex; at the DISC vertex these are the invasion payoffs of Supplementary Note 7. At an equilibrium on an edge, the tangent eigenvalue describes stability within the edge and the transverse eigenvalue is the growth rate of the absent strategy. Interior equilibria are classified by the Jacobian of Eq. (S26) on the simplex. Because reputations can jump where the selected branch ends, the replicator field can be discontinuous at folds (Fig. 3 and Extended Data Fig. 4).

Proposition S6 (Neutral line for Scoring without inertia)**.** *Under Scoring with $\mathscr{R}(u) = u$, $\Pi_{\mathrm{ALLC}} = \Pi_{\mathrm{ALLD}} = \Pi_{\mathrm{DISC}}$ at every composition with $f_{\mathrm{DISC}} = c/[b(\varepsilon - e_2)]$.*

Proof. Under SC, $\Phi_{\mathrm{C}} \equiv \varepsilon$ and $\Phi_{\mathrm{D}} \equiv e_2$ do not depend on the recipient, so $u_{\mathrm{DISC}} = \varepsilon(h_{\mathrm{GG}} + h_{\mathrm{GB}}) + e_2(h_{\mathrm{GB}} + h_{\mathrm{BB}}) = e_2 + (\varepsilon - e_2)\bar{r}$, and $r_{\mathrm{ALLC}} = \varepsilon$, $r_{\mathrm{ALLD}} = e_2$, $r_{\mathrm{DISC}} = e_2 + (\varepsilon - e_2)\bar{r}$. Then $\Pi_{\mathrm{ALLC}} - \Pi_{\mathrm{DISC}} = (1-e_1)(1-\bar{r})[b f_{\mathrm{DISC}}(\varepsilon - e_2) - c]$ and $\Pi_{\mathrm{ALLD}} - \Pi_{\mathrm{DISC}} = (1-e_1)\bar{r}[c - b f_{\mathrm{DISC}}(\varepsilon - e_2)]$, which both vanish at the stated f_{DISC}. □

At the reference parameters, $f_{\mathrm{DISC}} = 1/(5 \times 0.9408) \approx 0.2126$. This line of equilibria, neutral along its direction, is the dashed line in Fig. 3a (top). Trajectories that reach it drift neutrally along it, which is why integration under SC without inertia sometimes reaches the time cap without affecting the outcome class.

6.3 Cooperation measures

At a composition, intended and realized cooperation are ρ_{int} and ρ; in a homogeneous DISC population, $\rho_{\text{int}} = r_{\text{R}}^*$. To compare settings globally, trajectories are integrated from n_{init} area-uniform initial compositions. Across all scans, every final state belonged to one of two classes: the zero-cooperation set, comprising the ALLD vertex together with the low-reputation segment of the ALLD–DISC edge, on which DISC regard one another as bad, earn slightly less than ALLD and drift slowly towards ALLD (final $\rho \leq 7 \times 10^{-4}$); or an attractor with positive cooperation (final $\rho \geq 0.34$; between 0.974 and 0.980 in nearly all conditions). The cooperative basin is the set of initial compositions whose trajectories reach the second class. A trajectory is classified as cooperative if its final $\rho \geq 0.01$; every threshold in the observed gap gives the same classification, and the gap was checked for every condition. With $\bar{\rho}_{\text{D}}$ the mean final cooperation of the remaining trajectories,

$$\bar{\rho} = \omega_{\text{C}}\,\bar{\rho}_{\text{C}} + (1-\omega_{\text{C}})\,\bar{\rho}_{\text{D}} \approx \omega_{\text{C}}\,\bar{\rho}_{\text{C}}, \tag{S27}$$

so a change in $\bar{\rho}$ can be attributed to the basin size and to the cooperation level within it. At the reference parameters, $\bar{\rho}_{\text{C}}$ is 0.975–0.980 for one-step and selected multi-step inertia under all norms, limited by implementation error, and at least 94% of the increase in $\bar{\rho}$ produced by the selected architectures comes from the enlarged basin. These quantities depend on the area-uniform sampling of initial compositions; they are not probabilities over any other distribution of starting points.

Supplementary Note 7. Invasion of a discriminator population by unconditional strategies

7.1 A rare mutant in a discriminator population

Consider a homogeneous DISC population whose reputation is the selected root r_{R}^*, and a rare mutant X that uses the residents' norm and record mechanism. Because X is rare, its recipients, its observers and the individuals it is assessed against are residents. Residents regard each other as good with probability r_{R}^*, and their assessment environment is unchanged.

Lemma S7 (Rare-mutant identity)**.** *Suppose that X intends to help a resident recipient with probability χ_{X}, independently of the observer's image of that recipient and independently across interactions. Then X is assessed positively with probability*

$$u_{\text{X}} = \chi_{\text{X}}\Phi_{\text{C}}^* + (1-\chi_{\text{X}})\Phi_{\text{D}}^* = u_{\text{R}}^* - (r_{\text{R}}^* - \chi_{\text{X}})\,\Delta\Phi^*, \tag{S28}$$

its stationary reputation is $r_{\text{X}}^* = \mathscr{R}(u_{\text{X}})$, and

$$\Lambda_{\text{X}} \equiv \frac{\Pi_{\text{X}} - \Pi_{\text{R}}}{c(1-e_1)} = \left(r_{\text{R}}^* - \chi_{\text{X}}\right) - \frac{b}{c}\left(r_{\text{R}}^* - r_{\text{X}}^*\right). \tag{S29}$$

Proof. The observer regards the resident recipient as good with probability r_{R}^*, independently of X's intention, so X's assessment probability is the mixture of Φ_{C}^* and Φ_{D}^* with weights χ_{X} and $1-\chi_{\text{X}}$. Independent actions imply independent assessments, so (A2) holds exactly for X's records in the fixed resident environment, and $r_{\text{X}}^* = \mathscr{R}(u_{\text{X}})$. A resident receives help from residents with probability r_{R}^* and helps with probability r_{R}^*: $\Pi_{\text{R}} = (1-e_1)(b-c)r_{\text{R}}^*$. The mutant receives help with probability r_{X}^* and helps with probability χ_{X}: $\Pi_{\text{X}} = (1-e_1)(br_{\text{X}}^* - c\chi_{\text{X}})$. Subtracting and dividing by $c(1-e_1) > 0$ gives Eq. (S29). The second form of Eq. (S28) uses $u_{\text{R}}^* = r_{\text{R}}^*\Phi_{\text{C}}^* + (1-r_{\text{R}}^*)\Phi_{\text{D}}^*$. □

The first term of Eq. (S29) is the cooperation cost the mutant saves relative to a resident; the second is the help it loses because it is regarded differently. The resident corresponds to $\chi = r_{\text{R}}^*$, ALLD to $\chi = 0$, ALLC to $\chi = 1$ and the opportunistic discriminator of Supplementary Note 8 to $\chi = (1-\delta)r_{\text{R}}^*$. All of these mutants therefore lie on the single curve $\chi \mapsto \mathscr{R}[u_{\text{R}}^* - (r_{\text{R}}^* - \chi)\Delta\Phi^*]$, which is $\mathscr{R}$ evaluated along the segment between Φ_{D}^* ($\chi = 0$) and Φ_{C}^* ($\chi = 1$), passing through u_{R}^* at $\chi = r_{\text{R}}^*$.

7.2 ALLD and ALLC

Write $r_{\text{ALLD}}^* = \mathscr{R}(\Phi_{\text{D}}^*)$ and $r_{\text{ALLC}}^* = \mathscr{R}(\Phi_{\text{C}}^*)$, with

$$\Phi_{\text{D}}^* = r_{\text{R}}^* P_{\text{D|G}}^+ + (1-r_{\text{R}}^*)P_{\text{D|B}}^+, \qquad \Phi_{\text{C}}^* = r_{\text{R}}^* P_{\text{C|G}}^+ + (1-r_{\text{R}}^*)P_{\text{C|B}}^+. \tag{S30}$$

These are the reputations of rare mutants in the resident population, not of homogeneous ALLC or ALLD populations; they equal r^*_{ALLC} and r^*_{ALLD} of the population equations evaluated at the DISC vertex. Lemma S7 gives

$$\lambda_{\mathrm{ALLD}} = (1-e_1)\big[c\,r^*_{\mathrm{R}} - b\,(r^*_{\mathrm{R}} - r^*_{\mathrm{ALLD}})\big], \qquad \lambda_{\mathrm{ALLC}} = (1-e_1)\big[b\,(r^*_{\mathrm{ALLC}} - r^*_{\mathrm{R}}) - c\,(1-r^*_{\mathrm{R}})\big], \tag{S31}$$

where $\lambda_X = \Pi_X - \Pi_{\mathrm{R}}$. ALLD saves the resident's donation cost cr^*_{R} but may be helped less; ALLC can be regarded better than residents because it also helps recipients that residents regard as bad, but pays for these donations.

Proposition S8 (Resistance to ALLC and ALLD)**.** *(a) If $r^*_{\mathrm{R}} > r^*_{\mathrm{ALLD}}$, rare ALLD is excluded ($\lambda_{\mathrm{ALLD}} < 0$) if and only if $b/c > R_{\mathrm{L}} = r^*_{\mathrm{R}}/(r^*_{\mathrm{R}} - r^*_{\mathrm{ALLD}})$. If $r^*_{\mathrm{R}} \le r^*_{\mathrm{ALLD}}$ and $r^*_{\mathrm{R}} > 0$, no b/c excludes ALLD.*

(b) If $r^*_{\mathrm{ALLC}} > r^*_{\mathrm{R}}$, rare ALLC is excluded ($\lambda_{\mathrm{ALLC}} < 0$) if and only if $b/c < R_{\mathrm{U}} = (1-r^*_{\mathrm{R}})/(r^*_{\mathrm{ALLC}} - r^*_{\mathrm{R}})$. If $r^*_{\mathrm{ALLC}} \le r^*_{\mathrm{R}} < 1$, ALLC is excluded for every b/c.

(c) If $r^*_{\mathrm{R}} > r^*_{\mathrm{ALLD}}$ and $r^*_{\mathrm{ALLC}} > r^*_{\mathrm{R}}$, the residents resist both mutants if and only if

$$\frac{r^*_{\mathrm{R}}}{r^*_{\mathrm{R}} - \mathscr{R}\big[r^*_{\mathrm{R}} P^+_{\mathrm{D|G}} + (1-r^*_{\mathrm{R}}) P^+_{\mathrm{D|B}}\big]} < \frac{b}{c} < \frac{1-r^*_{\mathrm{R}}}{\mathscr{R}\big[r^*_{\mathrm{R}} P^+_{\mathrm{C|G}} + (1-r^*_{\mathrm{R}}) P^+_{\mathrm{C|B}}\big] - r^*_{\mathrm{R}}}. \tag{S32}$$

Proof. Rearrange the signs of Eq. (S31): $\lambda_{\mathrm{ALLD}} < 0$ if and only if $b(r^*_{\mathrm{R}} - r^*_{\mathrm{ALLD}}) > cr^*_{\mathrm{R}}$, and $\lambda_{\mathrm{ALLC}} < 0$ if and only if $b(r^*_{\mathrm{ALLC}} - r^*_{\mathrm{R}}) < c(1-r^*_{\mathrm{R}})$. Divide by the positive quantities stated in each case. □

7.3 When is the window open?

Proposition S9 (Chord condition)**.** *Under the conditions of Proposition S8(c), the interval in Eq. (S32) is non-empty if and only if*

$$r^*_{\mathrm{R}}\,\mathscr{R}(\Phi^*_{\mathrm{C}}) + (1-r^*_{\mathrm{R}})\,\mathscr{R}(\Phi^*_{\mathrm{D}}) < \mathscr{R}(u^*_{\mathrm{R}}), \qquad u^*_{\mathrm{R}} = r^*_{\mathrm{R}}\Phi^*_{\mathrm{C}} + (1-r^*_{\mathrm{R}})\Phi^*_{\mathrm{D}}. \tag{S33}$$

Proof. $R_{\mathrm{L}} < R_{\mathrm{U}}$ is equivalent to $r^*_{\mathrm{R}}(r^*_{\mathrm{ALLC}} - r^*_{\mathrm{R}}) < (1-r^*_{\mathrm{R}})(r^*_{\mathrm{R}} - r^*_{\mathrm{ALLD}})$, that is, $r^*_{\mathrm{R}} r^*_{\mathrm{ALLC}} + (1-r^*_{\mathrm{R}}) r^*_{\mathrm{ALLD}} < r^*_{\mathrm{R}}$. Self-consistency gives $r^*_{\mathrm{R}} = \mathscr{R}(u^*_{\mathrm{R}})$. □

The resident's own action is a mixture of the actions of ALLC and ALLD with weights r^*_{R} and $1-r^*_{\mathrm{R}}$, and its assessment probability u^*_{R} is the corresponding mixture of Φ^*_{C} and Φ^*_{D}. The window is open precisely when the reputation of this mixture exceeds the same mixture of the two pure reputations, that is, when $\mathscr{R}$ lies above its chord between Φ^*_{D} and Φ^*_{C} at u^*_{R}.

Corollary S10 (No window without inertia)**.** *If $\mathscr{R}(u) = u$ and $\Delta\Phi^* > 0$, then $R_{\mathrm{L}} = R_{\mathrm{U}} = 1/\Delta\Phi^*$: resistance to ALLD requires $b/c > 1/\Delta\Phi^*$ and resistance to ALLC requires $b/c < 1/\Delta\Phi^*$, so no benefit-to-cost ratio gives resistance to both.*

Proof. With $\mathscr{R}(u) = u$, $r^*_{\mathrm{R}} = u^*_{\mathrm{R}}$, $r^*_{\mathrm{R}} - r^*_{\mathrm{ALLD}} = r^*_{\mathrm{R}}\Delta\Phi^*$ and $r^*_{\mathrm{ALLC}} - r^*_{\mathrm{R}} = (1-r^*_{\mathrm{R}})\Delta\Phi^*$. □

If $\Delta\Phi^* \le 0$ without inertia (SJ, where the instantaneous resident has $r^*_{\mathrm{R}} = 1/2$ and $\Delta\Phi^* = 0$), ALLD is regarded at least as well as the residents and cannot be excluded. The thresholds also have a marginal interpretation. Define the reputation lost per unit of withheld cooperation, $H_{\mathrm{ALLD}} = (r^*_{\mathrm{R}} - r^*_{\mathrm{ALLD}})/r^*_{\mathrm{R}}$, and the reputation gained per unit of extra cooperation, $H_{\mathrm{ALLC}} = (r^*_{\mathrm{ALLC}} - r^*_{\mathrm{R}})/(1-r^*_{\mathrm{R}})$. Then $R_{\mathrm{L}} = 1/H_{\mathrm{ALLD}}$, $R_{\mathrm{U}} = 1/H_{\mathrm{ALLC}}$, and the window is

$$H_{\mathrm{ALLC}} < \frac{c}{b} < H_{\mathrm{ALLD}}: \tag{S34}$$

cooperating more than the residents must not pay off, and cooperating less must not pay off. Because $\Phi^*_{\mathrm{C}} - u^*_{\mathrm{R}} = (1-r^*_{\mathrm{R}})\Delta\Phi^*$ and $u^*_{\mathrm{R}} - \Phi^*_{\mathrm{D}} = r^*_{\mathrm{R}}\Delta\Phi^*$, H_{ALLC} and H_{ALLD} are $\Delta\Phi^*$ times the slopes of the chords of $\mathscr{R}$ on $[u^*_{\mathrm{R}}, \Phi^*_{\mathrm{C}}]$ and $[\Phi^*_{\mathrm{D}}, u^*_{\mathrm{R}}]$, respectively. Inertia opens the window by making $\mathscr{R}$ flatter above the resident's assessment probability than below it, so that extra cooperation buys little additional reputation while withheld cooperation is still penalized. Supplementary Table S7 lists the values at the reference parameters; Supplementary Note 11 gives the numerical classification rule.

Table S7. Reference values for a resident discriminator population. Highest stable homogeneous-DISC root r_{R}^*, reputations of rare ALLD and ALLC, and the thresholds of Eq. (S32); $e_1 = e_2 = 0.02$, $\alpha = \beta = 1$ for inertial settings. Without inertia $R_{\mathrm{L}} = R_{\mathrm{U}}$ (Corollary S10); under SJ without inertia ALLD is regarded as well as residents and cannot be excluded. Values agree with the verified data to all digits shown.

Norm	Setting	r_{R}^*	r_{ALLD}^*	r_{ALLC}^*	R_{L}	R_{U}
SC	none $(0,0)$	0.337838	0.020000	0.960800	1.0629	1.0629
	one-step $(1,1)$	0.996279	0.000808	0.996869	1.0008	6.3070
	selected $(2,0)$	0.999940	0.058808	0.999940	1.0625	230.24
SH	none $(0,0)$	0.020391	0.020000	0.039184	52.127	52.127
	one-step $(1,1)$	0.995272	0.000808	0.996111	1.0008	5.6409
	selected $(3,0)$	0.999998	0.077632	0.999998	1.0842	4410.4
SJ	none $(0,0)$	0.500000	0.500000	0.500000	–	–
	one-step $(1,1)$	0.995307	0.001215	0.996133	1.0012	5.6865
	selected $(4,3)$	0.9999996	8.3×10^{-7}	0.9999996	1.0000	20910
SS	none $(0,0)$	0.833333	0.180000	0.964000	1.2755	1.2755
	one-step $(1,1)$	0.996297	0.001122	0.996880	1.0011	6.3441
	selected $(1,2)$	0.993660	0.000036	0.995245	1.0000	4.0001

Supplementary Note 8. Opportunistic invasion and secure readout

8.1 The opportunistic discriminator

An opportunistic discriminator (O-DISC) uses the residents' norm and record mechanism and follows DISC, except that whenever its own image of the recipient is G it independently replaces the intended cooperation by intended defection with probability $\delta \in [0,1]$. This choice precedes implementation error, and O-DISC never cooperates with recipients it regards as bad. Its images of resident recipients are good with probability r_{R}^* and are independent of the observer's images by (A1), so it intends to help with probability $\chi = (1-\delta) r_{\mathrm{R}}^*$. By Lemma S7,

$$u_{\mathrm{X}}(\delta) = (1-\delta) r_{\mathrm{R}}^* \Phi_{\mathrm{C}}^* + \big[1 - (1-\delta) r_{\mathrm{R}}^*\big] \Phi_{\mathrm{D}}^* = u_{\mathrm{R}}^* - \delta\, r_{\mathrm{R}}^* \Delta\Phi^*, \qquad r_{\mathrm{X}}^*(\delta) = \mathscr{R}\big[u_{\mathrm{X}}(\delta)\big], \tag{S35}$$

$$\Lambda(\delta) = \frac{\Pi_{\mathrm{X}} - \Pi_{\mathrm{R}}}{c(1-e_1)} = \delta\, r_{\mathrm{R}}^* - \frac{b}{c}\big[r_{\mathrm{R}}^* - r_{\mathrm{X}}^*(\delta)\big]. \tag{S36}$$

Equivalently, $u_{\mathrm{X}}(\delta) = (1-\delta) u_{\mathrm{R}}^* + \delta \Phi_{\mathrm{D}}^*$: the mutant's assessment probability moves linearly from the resident's ($\delta = 0$) to that of ALLD ($\delta = 1$). Hence $\Lambda(0) = 0$, the mutant being a neutral copy of a resident, and $\Lambda(1) = \lambda_{\mathrm{ALLD}} / [c(1-e_1)]$. The family contains neither ALLC nor mutants whose omissions are correlated in time.

8.2 Weak opportunism

Differentiating Eq. (S36), with $\mathrm{d}r_{\mathrm{X}}^*/\mathrm{d}\delta = -r_{\mathrm{R}}^* \Delta\Phi^* \mathscr{R}'[u_{\mathrm{X}}(\delta)]$,

$$\Lambda'(0) = r_{\mathrm{R}}^* \Big[1 - \frac{b}{c} \Delta\Phi^* \mathscr{R}'(u_{\mathrm{R}}^*)\Big]. \tag{S37}$$

If $\Lambda'(0) > 0$, all sufficiently weak opportunists invade; if $\Lambda'(0) < 0$, they are excluded, but larger δ may still invade. The product $\Delta\Phi^* \mathscr{R}'(u_{\mathrm{R}}^*)$ is the reputation lost per unit of withheld cooperation for an infinitesimal deviation: the norm converts withheld cooperation into a lower positive-assessment probability at rate $\Delta\Phi^*$, and the record converts this into a lower reputation at rate $\mathscr{R}'(u_{\mathrm{R}}^*)$.

8.3 Resistance to the whole family

For $r_{\mathrm{R}}^* > 0$ and $0 < \delta \le 1$, define the reputation lost per unit of withheld cooperation,

$$H(\delta) = \frac{r_{\mathrm{R}}^* - r_{\mathrm{X}}^*(\delta)}{\delta\, r_{\mathrm{R}}^*} = \Delta\Phi^* \frac{\mathscr{R}(u_{\mathrm{R}}^*) - \mathscr{R}[u_{\mathrm{X}}(\delta)]}{u_{\mathrm{R}}^* - u_{\mathrm{X}}(\delta)} = \Delta\Phi^* \int_0^1 \mathscr{R}'\big(u_{\mathrm{R}}^* - s\,\delta\, r_{\mathrm{R}}^* \Delta\Phi^*\big)\, \mathrm{d}s, \tag{S38}$$

and its continuous extension $H(0) = \Delta\Phi^* \mathscr{R}'(u_R^*)$. The second form uses $r_R^* = \mathscr{R}(u_R^*)$ and $u_R^* - u_X = \delta r_R^* \Delta\Phi^*$ (for $\Delta\Phi^* \neq 0$): $H(\delta)$ is $\Delta\Phi^*$ times the slope of the chord of $\mathscr{R}$ between $u_X(\delta)$ and u_R^*. Then $\Lambda(\delta) = \delta r_R^*[1-(b/c)H(\delta)]$, and $H(1) = H_{\mathrm{ALLD}}$.

Proposition S11 (Family criterion)**.** *Let $r_R^* > 0$ and $H_{\min} = \min_{0\le\delta\le 1} H(\delta)$. Then $\Lambda(\delta) \le 0$ for every $\delta \in (0,1]$ if and only if $H_{\min} \ge c/b$. If $H_{\min} > c/b$, every O-DISC mutant is strictly excluded; if $H_{\min} < c/b$, a mutant with δ in a non-empty interval invades.*

Proof. For $\delta > 0$, $\Lambda(\delta)$ has the sign of $1-(b/c)H(\delta)$. The minimum includes $\delta = 0$ through the continuous extension, which is required because $\Lambda'(0)$ has the sign of $1-(b/c)H(0)$; if $H(0) < c/b$, all small $\delta > 0$ invade although no single $\delta > 0$ attains $H(0)$. Continuity of H gives the interval. □

Corollary S12 (Curvature and the location of the worst mutant)**.** *Let $\Delta\Phi^* > 0$.*

(a) If $\mathscr{R}$ is concave on $[\Phi_D^*, u_R^*]$, then H is non-decreasing in δ, $H_{\min} = H(0)$, and the family is excluded if and only if $\Lambda'(0) \le 0$. This holds for every architecture with $k_2 = 0$ (Proposition S4).

(b) If $\mathscr{R}$ is convex on $[\Phi_D^*, u_R^*]$, then $H_{\min} = H(1)$ and resistance to ALLD implies resistance to the whole family.

(c) In general, the worst mutant can be interior, and all δ must be checked.

Proof. As δ increases, $u_X(\delta)$ decreases from u_R^* to Φ_D^*. For concave $\mathscr{R}$, the slope of the chord from u_X to the fixed right endpoint u_R^* is non-increasing in u_X, hence non-decreasing in δ; for convex $\mathscr{R}$ it is non-increasing in δ. □

Part (a) also shows how tolerance creates vulnerability. With a concave response, the reputation lost by a mutant is at most the linear interpolation between the resident and ALLD, $r_R^* - r_X^*(\delta) \le \delta(r_R^* - r_{\mathrm{ALLD}}^*)$, so $\Lambda(\delta) \ge \delta\Lambda(1)$. Excluding ALLD then says little about partial opportunism. Near an all-good resident, Eq. (S16) gives $\mathscr{R}'(u_R^*) \approx 1-\alpha$ for $k_1 \ge 1$; with full tolerance readout, Eq. (S17) makes $\mathscr{R}'(u_R^*)$ of order $(1-u_R^*)^{k_1}$, so $H(0)$ and the marginal penalty vanish as the warning path lengthens. For SJ $(4,3)$ with $\beta = 1$ at the reference parameters, $H(0) = 4.8\times 10^{-5}$ at $\alpha = 1$ but 0.2037 at $\alpha = 0.8$, where $c/b = 0.2$.

8.4 An exact threshold in the error-free limit

Proposition S13 (Error-free security threshold)**.** *Let $e_1 = e_2 = 0$, $\beta = 1$ and $k_1 \ge 1$, and consider the all-good resident $r_R^* = 1$ under any of the four norms and any $k_2 \ge 0$. Every O-DISC mutant is excluded, $\Lambda(\delta) \le 0$ for all $\delta \in (0,1]$, if and only if*

$$\alpha \le 1 - \frac{c}{b}. \tag{S39}$$

Proof. Without errors, $P_{C|G}^+ = 1$ and $P_{D|G}^+ = 0$, so $r = 1$ is a resident root with $u_R^* = 1$ and $\Delta\Phi^* = 1$, and $u_X(\delta) = 1-\delta$. *Necessity.* By Eq. (S16), $\mathscr{R}'(1) = 1-\alpha$, so $\Lambda'(0) = 1-(b/c)(1-\alpha)$, which is positive if $\alpha > 1-c/b$. *Sufficiency.* With $\beta = 1$ the recovery term of Eq. (S10) vanishes, and for $0 < u \le 1$

$$\frac{\mathscr{R}_{k_1,k_2}(u)}{\mathscr{R}_{k_1,0}(u)} = \frac{u^{k_2}}{u^{k_2} + v^{k_1}(1-u^{k_2})} \le 1,$$

where subscripts denote depths. $\mathscr{R}_{k_1,0}$ is concave (Proposition S4) with $\mathscr{R}_{k_1,0}(1) = 1$ and $\mathscr{R}'_{k_1,0}(1) = 1-\alpha$, so it lies below its tangent at $u = 1$: $\mathscr{R}_{k_1,k_2}(u) \le \mathscr{R}_{k_1,0}(u) \le 1-(1-\alpha)(1-u)$. Hence $\Lambda(\delta) = \delta - (b/c)[1-\mathscr{R}_{k_1,k_2}(1-\delta)] \le \delta[1-(b/c)(1-\alpha)] \le 0$ when $\alpha \le 1-c/b$. □

At $b/c = 5$ the threshold is $\alpha = 0.8$, the readout used for illustration in Fig. 5 and Extended Data Fig. 9. With errors, the resident is no longer exactly all-good, $\Delta\Phi^* < 1$, and the boundary moves (for example to $\alpha_U = 0.8036$ under SJ $(4,3)$). The result is conditional on the all-good resident being selected. Stability of $r = 1$ itself requires $F'(1) = (1-\alpha)(2-p) - 1 < 0$, that is $\alpha > 0$ for SC and SS and $\alpha > 1/2$ for SH and SJ.

8.5 Secure readout intervals

Fix the norm, depths, β, errors and b/c. A readout α is *secure* if the high-reputation resident exists as a strictly stable root ($F=0$, $F'<0$) and satisfies $(b/c)H_{\min}\geq 1$. The secure set can be empty, can be the whole of $[0,1]$, or can consist of intervals; its boundaries are of three kinds.

1. *Fold (lower boundary α_{L}).* The high-reputation branch ends where $F(r,\alpha)=0$ and $\partial_r F(r,\alpha)=0$, with $F(r,\alpha)=\mathscr{R}[u_{\mathrm{R}}(r);\alpha,\beta,k_1,k_2]-r$, equivalently $\mathscr{R}'[u_{\mathrm{R}}(r)]\,u_{\mathrm{R}}'(r)=1$. The fold is marginally stable and is excluded. Because these equations contain no payoff parameter, α_{L} does not depend on b/c. By Eq. (S24), the resident reputation increases with α along the stable branch, and the branch is lost when α is lowered through α_{L}; in all cases computed, the resident then falls to a low-reputation root that opportunists invade.

2. *Invasion (upper boundary α_{U}).* $(b/c)H_{\min}=1$ on the high branch. In every case we computed with an interior upper boundary, the minimum was attained at $\delta\to 0$, so that

$$F(r,\alpha_{\mathrm{U}})=0,\qquad \frac{b}{c}\,\Delta\Phi(r)\,\mathscr{R}'\big[u_{\mathrm{R}}(r);\alpha_{\mathrm{U}}\big]=1. \tag{S40}$$

   For $k_2=0$ this is exact by Corollary S12(a). The boundary is included (neutral mutants only).

3. *Domain.* $\alpha=0$ or $\alpha=1$; included.

Because the resident does not depend on b/c and the condition $(b/c)H_{\min}\geq 1$ is monotone in b/c, the secure set grows with b/c: upper boundaries rise with b/c, whereas fold boundaries stay fixed (Extended Data Fig. 9b–e).

Proposition S14 (Where the secure interval closes)**.** *Suppose that near a fold $(r_{\mathrm{L}},\alpha_{\mathrm{L}})$ of the high branch the upper boundary is determined by Eq. (S40). Then an interval of secure readouts adjoins the fold if $b/c>u_{\mathrm{R}}'(r_{\mathrm{L}})/\Delta\Phi(r_{\mathrm{L}})$ and not if $b/c<u_{\mathrm{R}}'(r_{\mathrm{L}})/\Delta\Phi(r_{\mathrm{L}})$. Under Shunning this critical ratio is exactly 2 for every error rate, depth and readout.*

Proof. At the fold, $\mathscr{R}'u_{\mathrm{R}}'=1$, so the local security condition $(b/c)\Delta\Phi\mathscr{R}'\geq 1$ reads $b/c\geq u_{\mathrm{R}}'/\Delta\Phi$. Under SH, $u_{\mathrm{R}}(r)=e_2+\kappa r^2$ and $\Delta\Phi(r)=\kappa r$ (Supplementary Table S4), so $u_{\mathrm{R}}'/\Delta\Phi=2\kappa r/(\kappa r)=2$. □

For SJ, $u_{\mathrm{R}}'(r)/\Delta\Phi(r)=2-e_1(1-2e_2)/\Delta\Phi(r)$, which is 1.9710 at the fold of the selected $(4,3)$ architecture; for SC, $u_{\mathrm{R}}'/\Delta\Phi=1$, so a fold never closes the interval for $b/c>1$. These results explain why the secure intervals of SH close at $b/c=2$ in Extended Data Fig. 9c, whereas that of the selected SJ architecture is still open, though narrow, at $b/c=2$.

8.6 Worked example: Stern Judging with depths (4,3)

At $e_1=e_2=0.02$: $P^{+}_{\mathrm{C|G}}=0.9608$, $P^{+}_{\mathrm{D|G}}=0.02$, $P^{+}_{\mathrm{C|B}}=0.0392$ and $P^{+}_{\mathrm{D|B}}=0.98$, so $\Delta\Phi(r)=0.9408(2r-1)$ and $u_{\mathrm{R}}(r)=0.98-1.9008\,r+1.8816\,r^2$. With $\beta=1$,

$$\mathscr{R}(u;\alpha)=\frac{u^4\sum_{m=0}^{4}[\alpha(1-u)]^m}{(1-u)^4+u^3-(1-u)^4u^3}.$$

- $\alpha=1$: $r_{\mathrm{R}}^{*}=0.9999996$; ALLD is excluded ($\Lambda(1)=-4.000$), but $\Lambda(\delta)>0$ for $0<\delta<\delta_{\mathrm{c}}=0.365299$.
- $\alpha=0.8$: $r_{\mathrm{R}}^{*}=0.986509$ and $H_{\min}=H(0)=0.20366>c/b$; every O-DISC mutant is excluded, with $(b/c)_{\mathrm{crit}}=1/H_{\min}=4.910$.
- Fold: $\alpha_{\mathrm{L}}=0.5669858$ at $r=0.851537$. Upper boundary: $\alpha_{\mathrm{U}}=0.8036040$ at $r=0.986906$, where $F'(r)<0$. The secure interval is $0.5670<\alpha\leq 0.8036$.

The trade-off at $\delta=0.1$ illustrates the mechanism:

Readout	r_{R}^{*}	$r_{\mathrm{X}}^{*}(0.1)$	Saved cost $\delta r_{\mathrm{R}}^{*}$	Lost benefit $(b/c)(r_{\mathrm{R}}^{*}-r_{\mathrm{X}}^{*})$	$\Lambda(0.1)$
$\alpha=1$	0.9999996	0.999789	0.100000	0.001054	+0.098946
$\alpha=0.8$	0.986509	0.964412	0.098651	0.110484	−0.011833

Supplementary Table S8 lists the secure intervals of all architectures at $b/c=5$.

Table S8. Secure readout intervals at $b/c = 5$ ($\beta = 1$, $e_1 = e_2 = 0.02$). Round brackets: excluded fold endpoints; square brackets: included domain or neutral-invasion endpoints. Last two columns: resident reputation and the critical benefit-to-cost ratio $1/H_{\min}$ at $\alpha = 0.8$ for the selected architectures. One-step SH and SJ have lost the high-reputation resident at $\alpha = 0.8$.

Norm	One-step $(1,1)$	Selected multi-step	$r_{\mathrm{R}}^*(0.8)$	$(b/c)_{\mathrm{crit}}(0.8)$
SC	$(0.431929, 0.966915]$	$(2,0)$: $[0, 0.807064]$	0.989782	4.827
SH	$(0.864863, 0.989948]$	$(3,0)$: $(0.591161, 0.806464]$	0.986341	4.835
SJ	$(0.852446, 0.988514]$	$(4,3)$: $(0.566986, 0.803604]$	0.986509	4.910
SS	$[0, 0.965245]$	$(1,2)$: $[0, 1]$	0.977886	1.955

Supplementary Note 9. Alternative reputation filters

Fig. 5a compares the inertial record with two mechanisms that also buffer isolated negative evidence. In both cases only the conversion of assessments into a binary image changes: the norm, errors, the private-assessment closure, the resident selection (highest root reached from $r = 1$) and Eqs. (S35)–(S36) with $\mathscr{R}$ replaced by the alternative map are kept. The original studies use different strategy sets, observation protocols and evolutionary dynamics, so these are controlled adaptations, not replications; their parameters were neither optimized nor matched in memory to the inertial record.

9.1 Aggregation

Following ref. [5], an observer samples M assessments of the donor and assigns a good image if at least Q of them are positive. If each sampled assessment is independently positive with probability u, the number of positive ones is binomial, and

$$\mathscr{R}_{\mathrm{agg}}(u;M,Q) = \sum_{j=Q}^{M} \binom{M}{j} u^j (1-u)^{M-j}, \qquad \mathscr{R}'_{\mathrm{agg}}(u) = M \binom{M-1}{Q-1} u^{Q-1}(1-u)^{M-Q}. \tag{S41}$$

We use $(M,Q) = (2,1)$, the rule that forgives one of two negative assessments: $\mathscr{R}_{\mathrm{agg}}(u) = 1-(1-u)^2 = 2u-u^2$. Then $r_{\mathrm{R}}^* - r_{\mathrm{X}}^* = 2(1-u_{\mathrm{R}}^*)x + x^2$ with $x = \delta r_{\mathrm{R}}^* \Delta\Phi^*$, and

$$\Lambda_{\mathrm{agg}}(\delta) = r_{\mathrm{R}}^* \, \delta \left[1 - 2\frac{b}{c}(1-u_{\mathrm{R}}^*)\Delta\Phi^* - \frac{b}{c} r_{\mathrm{R}}^* \, (\Delta\Phi^*)^2 \, \delta \right]. \tag{S42}$$

For SJ at the reference parameters, $r_{\mathrm{R}}^* = 0.998188$, $u_{\mathrm{R}}^* = 0.957431$ and $\Delta\Phi^* = 0.937390$, giving $\Lambda_{\mathrm{agg}} = 0.599876\,\delta - 4.377595\,\delta^2$ and invasion for $0 < \delta < 0.137033$. In the error-free all-good limit, $\Lambda_{\mathrm{agg}} = \delta[1-(b/c)\delta]$ and the interval ends at $\delta = c/b$.

9.2 Quantitative score

Following ref. [7], an observer keeps an integer score $z \in \{-z_{\max}, \ldots, z_{\max}\}$ for each target, increases it by one after a positive and decreases it by one after a negative assessment (attempts to leave the range leave the score unchanged), and reads the target as good if $z \geq z_0$. For i.i.d. input $u \in (0,1)$, the score is a reflecting birth–death chain on $j = z + z_{\max} \in \{0, \ldots, 2z_{\max}\}$. Detailed balance, $\pi_j \, u = \pi_{j+1}(1-u)$, gives $\pi_j \propto t^j$ with $t = u/(1-u)$, and

$$\mathscr{R}_{\mathrm{score}}(u; z_{\max}, z_0) = \frac{\sum_{j=z_{\max}+z_0}^{2z_{\max}} t^j}{\sum_{j=0}^{2z_{\max}} t^j}, \qquad \mathscr{R}_{\mathrm{score}}(0) = 0, \; \mathscr{R}_{\mathrm{score}}(1) = 1. \tag{S43}$$

At $u = 1/2$ the score is uniform and $\mathscr{R}_{\mathrm{score}} = (z_{\max} - z_0 + 1)/(2z_{\max}+1)$; for $z_{\max} = 1$, $z_0 = 0$, $\mathscr{R}_{\mathrm{score}} = u/(1-u+u^2)$. We use $z_{\max} = 5$, $z_0 = 0$. Under SJ at the reference parameters the selected resident has $1 - r_{\mathrm{R}}^* \approx 4.6 \times 10^{-9}$, ALLD is excluded, and O-DISC invades for $0 < \delta < 0.383206$. The source model starts scores at zero; selecting the root reached from $r = 1$ is a convention of this adaptation, and SC and SH also have lower roots for $z_{\max} = 5$.

Supplementary Table S9 summarizes the five curves of Fig. 5a.

Table S9. Opportunistic invasion under SJ ($b/c = 5$, $e_1 = e_2 = 0.02$, $\beta = 1$). δ_c: upper end of the interval with $\Lambda > 0$ (0 if none).

Mechanism	r_R^*	$\Lambda(1)$	δ_c
One-step inertia $(1,1)$, $\alpha = 1$	0.995307	−3.975	0.008397
Selected $(4,3)$, $\alpha = 1$	0.9999996	−4.000	0.365299
Selected $(4,3)$, $\alpha = 0.8$	0.986509	−3.946	0
Aggregation, $(M,Q) = (2,1)$	0.998188	−3.778	0.137033
Quantitative score, $z_{max} = 5$, $z_0 = 0$	0.99999999	−4.000	0.383206

Supplementary Note 10. Finite-population reputation simulations

10.1 Purpose and scope

The simulations test how reputations form when the mean-field closure is not imposed. Strategy counts are fixed throughout every run: the simulations concern reputation dynamics only and contain no strategy replacement, mutation or imitation. Strategy evolution is analysed with the replicator dynamics of Supplementary Note 6. The simulations retain individual latent records, sampling fluctuations, differences among targets of the same strategy and correlations between the records of different observers.

10.2 Event algorithm

The population has N individuals with fixed counts n_{ALLC}, n_{ALLD} and n_{DISC}. Each DISC individual i stores a latent record L_{ij} for every individual j, including itself; all records start in G. Records of ALLC and ALLD individuals are not stored: these strategies never consult reputations, and observers are drawn from the DISC individuals, so omitting them changes no event. Each event consists of the following steps.

1. Draw a donor uniformly from the N individuals and a recipient uniformly from the other $N-1$. Draw an observer uniformly from the n_{DISC} DISC individuals, independently of donor and recipient; the observer may coincide with either.
2. Determine the intended action: C for ALLC, D for ALLD; a DISC donor samples an image from its record of the recipient, Bernoulli$[\sigma(L_{donor,recipient})]$, and intends C if the image is G.
3. If the intended action is C, replace it by D with probability e_1.
4. The observer samples its own image of the recipient, Bernoulli$[\sigma(L_{observer,recipient})]$, independently of step 2, and draws a prescribed assessment that is positive with probability ν_{AY} for the realized action A and its image Y. The assessment is reversed with probability e_2.
5. Apply T^+ or T^- (Supplementary Table S5) to $L_{observer,donor}$. No other record changes.

One sweep consists of $n_{DISC}N$ events, the number of stored records, so each record is updated once per sweep on average. When the DISC donor is itself the observer, its two consultations in steps 2 and 4 are independent draws from the same record, and the update occurs after both. The instantaneous panels of Fig. 2a–d use the one-step state space with $\alpha = \beta = 0$, which generates the same image process as $(k_1,k_2) = (0,0)$ (Lemma S1).

10.3 Estimators

At a sampled time, let $s_{ij} = \sigma(L_{ij})$, the probability that observer i currently reads target j as good. Using these probabilities, rather than sampled images, removes readout noise from the estimates. For target j, the mean goodness and the probability that two distinct DISC observers, consulting independently, read the target differently are

$$g_j = \frac{1}{n_{DISC}} \sum_i s_{ij}, \qquad d_j = \binom{n_{DISC}}{2}^{-1} \sum_{i<i'} \left[s_{ij}(1-s_{i'j}) + (1-s_{ij})s_{i'j} \right] = \frac{(n_{DISC}-1)\mu_{1,j} - \mu_{1,j}^2 + \mu_{2,j}}{\binom{n_{DISC}}{2}}, \tag{S44}$$

where $\mu_{1,j} = \sum_i s_{ij}$ and $\mu_{2,j} = \sum_i s_{ij}^2$, and i, i' range over the DISC observers (including j itself if j uses DISC). The second form follows from $\sum_{i<i'}(s_i + s_{i'} - 2s_i s_{i'}) = (n_{\mathrm{DISC}} - 1)\mu_1 - (\mu_1^2 - \mu_2)$. For deterministic readouts with a fraction x_j of observers reading j as good, $d_j = 2n_{\mathrm{DISC}} x_j(1-x_j)/(n_{\mathrm{DISC}} - 1)$, the finite-observer version of $2x_j(1-x_j)$. The strategy-level quantities g_I and d_I average g_j and d_j over the n_I targets of strategy I and over sampling times, and

$$\Delta g = \frac{g_{\mathrm{DISC}} + g_{\mathrm{ALLC}}}{2} - g_{\mathrm{ALLD}}, \qquad \bar{d} = \sum_I \frac{n_I}{N} d_I. \tag{S45}$$

Δg measures how differently cooperative strategies and unconditional defectors are treated; it is not a classification accuracy. Low $\bar{d}$ alone is not evidence of good reputation formation, because observers can agree that everyone is bad (one-step SH in Fig. 2b). The two statistics are therefore interpreted jointly. g_I is the finite-population counterpart of r_I, but it is estimated without the closure.

10.4 Runs, replication and uncertainty

- *Fig. 2a–d (image matrices).* $N = 90$ (30 per strategy); 1,000 sweeps; seed 2027 for each panel; the matrix shows the final readout probabilities s_{ij} of the DISC observers (rows) for all targets (columns, grouped by strategy), which are 0 or 1 at the displayed readouts. These are illustrative single realizations.
- *Fig. 2e–h (depth maps).* $N = 150$ (50 per strategy); $k_1, k_2 \in \{0, \ldots, 6\}$, $\alpha = \beta = 1$; one run per condition (base seed 2027); 1,000 sweeps, sampled every 5 sweeps from sweep 800 to 1,000 inclusive (41 samples). Colour shows Δg and circle area $\bar{d}$.
- *Extended Data Fig. 1.* The same 196 conditions with 30 independent runs each (5,880 runs). Condition seeds are generated by NumPy `SeedSequence` from the base seed 20260918, the norm index (SC, SH, SJ, SS), k_1, k_2 and the run index. Point estimates are means of the 30 run means; pointwise 95% percentile intervals use 2,000 bootstrap resamples of run means (seed 2026091801). Samples within a run are never treated as independent.
- *Drift criterion.* For each run, the difference between the means of the second and first halves of its sampling window is computed for Δg and $\bar{d}$, and these differences are bootstrapped across runs. A condition passes if both 95% intervals lie inside $(-0.01, 0.01)$. Conditions that failed were rerun for 3,000, 10,000 and then 50,000 sweeps, with the first 80% of each run discarded and the five-sweep interval retained, until they passed. Only SJ $(4,5)$ (3,000 sweeps) and SJ $(3,5)$ (50,000 sweeps) required extension, and all 196 conditions passed. This is a practical check of the sampling window, not a proof of mixing.

10.5 Selection of representative architectures

Among conditions that pass the drift criterion and satisfy $\Delta g \geq 0.8$ and $\bar{d} \leq 0.1$, we choose the smallest total depth K, then the larger Δg, the smaller $\bar{d}$, the smaller k_1 and the smaller k_2. This selects SC $(2,0)$, SH $(3,0)$, SJ $(4,3)$ and SS $(1,2)$ (Supplementary Table S10). The cutoffs are operational criteria, not theoretical thresholds, and selection uses no evolutionary or invasion outcome. The stars in Fig. 2e–h apply the same cutoffs and tie-breaking rule to the single-run maps (without the drift criterion) and select the same four pairs. Extended Data Fig. 1e–h repeats the selection for minimum separations $0.70, 0.75, \ldots, 0.90$ and maximum disagreements $0.050, 0.075, \ldots, 0.150$.

Table S10. Selected architectures (30 runs, $N = 150$; brackets: pointwise 95% bootstrap intervals).

Norm	(k_1, k_2)	Δg	$\bar{d}$
SC	$(2,0)$	0.9195 [0.9192, 0.9198]	0.0644 [0.0641, 0.0646]
SH	$(3,0)$	0.9059 [0.9056, 0.9062]	0.0694 [0.0691, 0.0696]
SJ	$(4,3)$	0.8806 [0.8801, 0.8812]	0.0812 [0.0809, 0.0815]
SS	$(1,2)$	0.9174 [0.9171, 0.9177]	0.0562 [0.0560, 0.0564]

10.6 Relation to the mean-field theory

The simulations and the theory share the event rules, records and readout, but differ in the closure, in population size and in that strategies do not evolve. The simulations are used to identify architectures that produce discriminating and consistent private reputations; the strategy dynamics then use the closed theory with these architectures. No quantitative agreement between simulated g_I and mean-field r_I is claimed or used.

Supplementary Note 11. Numerical methods

11.1 Reputation map and roots

$\mathscr{R}$ is evaluated from Eq. (S10), with arguments clipped to $[0,1]$ and the special case $\alpha=\beta=0$ evaluated as $\mathscr{R}(u)=u$; $\mathscr{R}'$ is evaluated from Eq. (S15) or by central differences. The closed form was checked against the stationary vector of $T(u)$ computed by linear algebra for random depths, readouts and inputs. Homogeneous-DISC roots are found by evaluating F on a uniform grid (2,001 points for the drift curves and the depth planes, 401 points for the 161×161 readout plane, 4,001 points in Fig. 5), refining every sign change by bracketed root finding and classifying roots as stable or unstable when $|F'|>10^{-7}$ and as near-fold otherwise. In Extended Data Fig. 9, candidate roots are the real roots of the polynomial numerator of F (from a Chebyshev fit) and are refined by Newton's method.

11.2 Reputations at mixed compositions

Eq. (S22) is integrated from $r_I=1$ by fourth-order Runge–Kutta steps, with every stage clipped to $[0,1]$. The strategy engine uses step 0.25 until $\max_I|\mathscr{R}(u_I)-r_I|\le10^{-11}$ (fallback step 0.1). The simplex fields of Fig. 3 use steps 0.1 (up to 2×10^4 steps) and then 0.2 (up to 8×10^4 and a further 2×10^5 steps if needed), with residual $\le10^{-12}$.

11.3 Residents for local invasion

The scalar flow $\mathrm{d}r/\mathrm{d}\tau=F(r)$ is integrated from $r=1$ by RK4 with step 0.18, to residual 10^{-11} (Extended Data Figs. 5, 7 and 8i–l) or 10^{-9} (Extended Data Fig. 6), and the root is then refined by Newton steps of at most 10^{-6} to residual $\le10^{-13}$. The step bound keeps the branch reached by relaxation. This refinement matters where the resident reached from all-good is the all-bad state ($e_2=0$ under SC and SH), at which ALLD is exactly neutral. Local outcomes are classified from λ_{ALLD} and λ_{ALLC} (with $c=1$): ALLD is not excluded if $\lambda_{\mathrm{ALLD}}\ge-10^{-10}$ (this class includes neutral ALLD); otherwise ALLC invades if $\lambda_{\mathrm{ALLC}}>10^{-10}$; otherwise the residents resist both. Dots in Extended Data Fig. 7 mark the last class, and solid segments in Extended Data Fig. 5 mark $R_{\mathrm{L}}<b/c<R_{\mathrm{U}}$.

11.4 Strategy trajectories

The engine tabulates the all-good-selected reputations exactly at the 7,381 nodes of a triangular lattice with 120 subdivisions, node (i,j) being $(f_{\mathrm{ALLC}},f_{\mathrm{ALLD}})=(i,j)/120$. A cell is flagged if any reputation component varies by more than 0.02 among its three nodes. In unflagged cells reputations are interpolated barycentrically; in flagged cells, which include every cell crossed by a fold or a steep transition, they are computed exactly by relaxation at the requested composition. Payoffs and the replicator field are computed from these reputations. Earlier pipelines interpolated the replicator field itself on a coarse mesh across folds, which biased sampled cooperation; this was corrected. Trajectories are integrated by the explicit midpoint rule with $\Delta t=0.2$, renormalizing $\boldsymbol{f}$ to the simplex after each stage. After every block of 500 steps (100 time units), ρ_{int} is compared with its value after the previous block; integration stops after three consecutive changes of at most 10^{-7}, or after 400 blocks (4×10^4 time units).

Validation. (i) Refined settings (160 subdivisions, flag threshold 0.01, $\Delta t=0.1$, 1,000-step blocks, tolerance 10^{-8}) changed $\bar{\rho}$ by at most 1.3×10^{-5} and no outcome class. (ii) An independent implementation that recomputes reputations exactly at every step, without a lattice, reproduced the outcome of every tested trajectory. (iii) The analytic SC no-inertia case is reproduced. (iv) The time cap was reached only on the neutral segment of SC without inertia (Proposition S6) and by a few trajectories at b/c close to R_{L} or R_{U}, where an invasion payoff is close to zero; the outcome-gap check (Supplementary Note 6) holds for every condition.

11.5 Initial compositions

Shifted Halton (Fig. 4; Extended Data Figs. 2e, 7, 8e–h): points $i = 1, \ldots, 512$ of the Halton sequence in bases 2 and 3, shifted modulo 1 by a uniform random vector drawn with seed 2026. *Scrambled Sobol* (Extended Data Fig. 5): eight independently scrambled two-dimensional Sobol sequences of 512 points (SciPy `qmc.Sobol` [8], seeds 2026–2033). A point (x, y) of the unit square with $x + y > 1$ is reflected to $(1 - x, 1 - y)$, which maps the square two-to-one onto the triangle with constant density, and $(f_{\mathrm{ALLC}}, f_{\mathrm{ALLD}}, f_{\mathrm{DISC}}) = (x, y, 1 - x - y)$. The same compositions are used for all conditions within a figure.

11.6 Opportunistic invasion

Fig. 5 evaluates $\Lambda(\delta)$ at 5,015 distinct values of δ: 4,001 uniform points on $[0, 1]$, 1,001 on $[0, 0.025]$ and 100 logarithmically spaced points on $[10^{-9}, 10^{-3}]$. Nonzero sign changes are refined by Brent's method (tolerance 2×10^{-13}). The readout scan uses 1,001 uniform values of α plus points next to the two boundaries (1,006 values). α_{L} is obtained by solving $F = \partial_r F = 0$ and α_{U} by Brent's method on Eq. (S40); the full δ-grid confirmed that no finite δ invades inside the interval. Residents of the alternative filters are obtained from $r = 1$ by $r \leftarrow 0.8r + 0.2\mathscr{R}_{\mathrm{X}}[u_{\mathrm{R}}(r)]$ until successive values differ by less than 2×10^{-14}.

Extended Data Fig. 9 uses at least 601 values of α, refined near folds, 161 values of $b/c \in [2, 8]$ and $\beta = 1$. Because $\mathscr{R}$ is rational in u and u_{X} is linear in δ, $H(\delta)$ is a rational function of δ. Its minimum is found by comparing $H(0)$, $H(1)$ and H at every real root in $(10^{-8}, 1)$ of the numerator of $\mathrm{d}H/\mathrm{d}\delta$ (roots accepted if their imaginary part is below 10^{-7}). A readout is secure if $1 - (b/c)H_{\min} \leq 10^{-10}$. This is a finite-precision search, not a proof that no extremely narrow interior extremum exists; in all reported cases the minimum was at $\delta \to 0$.

11.7 Software

Analyses use Python 3.11 or later with NumPy, SciPy, pandas and Matplotlib, a C99 implementation of the strategy engine, and Numba for the finite-population simulations and simplex fields. The verification was run with Python 3.13.17, NumPy 2.5.3, pandas 3.0.5, SciPy 1.18.1, Matplotlib 3.11.2 and GCC 13.3. Every notebook loads the supplied data by default and regenerates it when its recompute option is set.

Supplementary Note 12. Figure-by-figure settings

Supplementary Table S11 lists, for every main-text and Extended Data figure, the model settings, the quantity shown, the numerical procedure and the code. Unless stated, $b/c = 5$ and $e_1 = e_2 = 0.02$. Paths refer to the verified computational package (`verified/`); working numbers of Extended Data figures in that package differ from the manuscript numbering and are given in parentheses.

Table S11. Settings and procedures for each figure.

Figure	Model and quantity	Procedure and sample	Code
Fig. 1	Schematic of instantaneous and inertial updating and readout.	Conceptual; no data.	–
Fig. 2a–d	Private images s_{ij} of DISC observers; no inertia (one-step space, $\alpha = \beta = 0$), full one-step and selected multi-step inertia.	Finite simulation, $N = 90$ (30 per strategy), 1,000 sweeps, seed 2027; final state of one run.	`figure2_ed1/ Figure2.ipynb`
Fig. 2e–h	Δg (colour) and $\bar{d}$ (circle area) for $0 \leq k_1, k_2 \leq 6$, $\alpha = \beta = 1$.	$N = 150$, one run per condition (base seed 2027), samples every 5 sweeps from 800 to 1,000. Stars: selection rule without drift criterion.	`figure2_ed1/ Figure2.ipynb`
Fig. 3	Replicator dynamics; none, one-step, selected multi-step.	Field on 84×74 grid clipped to the simplex; reputations relaxed from all-good at every point (residual $\leq 10^{-12}$); vertex, edge (501-point scans) and interior equilibria; tangent stability for edge markers; strict tolerance 10^{-7}.	`figure3_ed2-ed5_ed9/ Figure3_Simplex_ Dynamics.ipynb`

Figure	Model and quantity	Procedure and sample	Code
Fig. 4	$\bar{\rho}$ for all 180 allocations with $K \le 8$, $\alpha = \beta = 1$; envelope = largest $\bar{\rho}$ at each K.	Engine; 512 shifted-Halton compositions (seed 2026).	`figure4_ed6/Figure4_Total_Depth.ipynb`; `engine/scan.py depth`
Fig. 5a	$\Lambda(\delta)$ under SJ: one-step; $(4,3)$ with $\alpha = 1$ and 0.8; aggregation $(2,1)$; score $z_{\max} = 5$, $z_0 = 0$; $\beta = 1$.	Highest strictly stable resident (4,001-point grid, Brent); 5,015 values of δ; Brent refinement of crossings; filters' residents by damped iteration.	`figure5_ed7/Figure5_Opportunistic_Invasion.ipynb`
Fig. 5b	δ_c and r_R^* against α, SJ $(4,3)$, $\beta = 1$.	1,006 values of α; α_L from $F = \partial_r F = 0$; α_U from Eq. (S40), checked on the full δ-grid.	as above
ED Fig. 1 (ED1)	Δg versus $\bar{d}$; selection under alternative cutoffs.	$N = 150$, 30 runs per condition (base seed 20260918), bootstrap (2,000 resamples, seed 2026091801), drift criterion; extended runs for SJ $(4,5)$, $(3,5)$.	`figure2_ed1/Extended_Data_Figure1.ipynb`
ED Fig. 2a–d (ED2)	Drift $F(r)$ of homogeneous DISC; none, one-step, selected.	Closed-form F; roots from 2,001-point grids; stability from F'.	`figure3_ed2-ed5_ed9/Extended_Data_Figure2.ipynb`
ED Fig. 2e (ED2)	$\bar{\rho}$ from all-good versus all-bad initial reputations.	Engine, quasi-static tracking protocol; 512 shifted-Halton compositions.	`engine/initial_reputation.py`
ED Fig. 3 (ED9)	Highest stable homogeneous-DISC root; hatching = two stable roots.	a–d: $(1,1)$, 161×161 grid of (α,β), 401-point root grids; e–h: $\alpha = \beta = 1$, $0 \le k_1, k_2 \le 8$, 2,001-point root grids; full-state stability check of every depth-plane root.	`figure3_ed2-ed5_ed9/Extended_Data_Figure2.ipynb`
ED Fig. 4 (ED8)	All self-consistent states on the ALLD–DISC edge; payoff advantage of DISC over ALLD on the all-good branch; folds.	Reduction to one equation in $\bar{r}$ (8,001-point grid, bisection) for 1,071 values of $f_{\mathrm{DISC}} \in [0.01, 1]$.	`ed8_reputation_geometry/Extended_Data_Figure8.ipynb`
ED Fig. 5 (ED4)	$\bar{\rho}$ against b/c; local protection segments; R_L, R_U.	Engine; 45 ratios; eight scrambled-Sobol samples of 512 (seeds 2026–2033); mean ± s.e. across samples; refined residents.	`figure3_ed2-ed5_ed9/Extended_Data_Figure4.ipynb`; `scan.py ratio`
ED Fig. 6 (ED5)	Local invasion classes over (e_1, e_2); none, one-step, selected.	241×241 grid on $[0, 0.45]^2$; residents relaxed to 10^{-9}, Newton-refined to 10^{-13}; classification rule of Note 11.	`figure3_ed2-ed5_ed9/Extended_Data_Figure5.ipynb`
ED Fig. 7 (ED6)	$\bar{\rho}$ for $0 \le k_1, k_2 \le 7$, $\alpha = \beta = 1$; dots: residents resist ALLD and ALLC.	Engine; 512 shifted-Halton compositions; dots from refined residents.	`figure4_ed6/Extended_Data_Figure6.ipynb`
ED Fig. 8a–d (ED3)	Replicator dynamics at $(1,1)$ with $(\alpha,\beta) = (1,0)$ and $(0,1)$.	As Fig. 3.	`Figure3_Simplex_Dynamics.ipynb`
ED Fig. 8e–h (ED3)	$\bar{\rho}$ over 41×41 grid of (α,β) at $(1,1)$.	Engine; 512 shifted-Halton compositions.	`scan.py readout`
ED Fig. 8i–l (ED3)	Local invasion classes over 801×801 grid of (α,β) at $(1,1)$.	Residents relaxed to 10^{-11} (polynomial fallback), Newton-refined to 10^{-13}.	`Extended_Data_Figure3_Planes.ipynb`
ED Fig. 9 (ED7)	Security against the O-DISC family across α and b/c; colour: r_R^*.	≥ 601 values of α (refined near folds), 161 values of $b/c \in [2,8]$, $\beta = 1$; rational-function minimization of H; tolerance 10^{-10}; fold endpoints excluded.	`figure5_ed7/Extended_Data_Figure7.ipynb`

Supplementary Note 13. Scope and limitations

1. *Closure.* The population-level results rest on assumptions (A1) and (A2). The single-record map $\mathscr{R}$ is exact for i.i.d. inputs, and the self-consistency equations are exact stationary equations of the closed latent-state dynamics (Proposition S5), but the closure itself is an approximation to the finite private-record process.

2. *Branch selection.* Several stationary reputation states can coexist. All strategy-dynamics results use the branch reached from good records; the dependence on this convention is examined in Extended Data Fig. 2e. Stability statements refer to the auxiliary flow; they agreed with the full latent-state linearization wherever this was checked, but the transient dynamics of the two descriptions differ.

3. *Timescales.* Strategy evolution is evaluated at stationary reputations; simultaneous reputation and strategy change is not modelled.

4. *Tested mutants.* Local resistance concerns rare ALLC, ALLD and the i.i.d. O-DISC family. It does not establish resistance to other strategies, including temporally correlated or persistent opportunism, and security against O-DISC does not imply resistance to ALLC (selected SS architecture).

5. *Sampled quantities.* $\bar{\rho}$ and ω_C refer to area-uniform initial compositions and to the stated numerical stopping rules; the uncertainty in Extended Data Fig. 5 describes the sampling of initial compositions only.

6. *Selection of architectures.* The four architectures are representative minimal-depth solutions of operational recognition criteria within $0 \leq k_1, k_2 \leq 6$, not global optima. The total-depth envelope is a maximum over the scanned allocations, not an optimization over unrestricted mechanisms, and inertia carries no explicit cost in the model.

7. *Alternative filters.* Aggregation and quantitative scoring are represented by single, unoptimized parameter settings and are not matched in memory; the comparison illustrates a shared vulnerability rather than ranking mechanisms.

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